

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

CHAPTER 1

INTRODUCTION

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Introduction

The introductory discussion also establishes the importance of looking at customers as individuals as well as members of broader groups. Individual customer responses may appear different when considered separately, but segmentation provides a method for organising those responses into meaningful categories. This is useful for the present study because the objective is to understand purchasing patterns through identifiable customer characteristics. The present study begins from the understanding that online shopping has changed the nature of the relationship between customers and fashion retailers. In a digital environment, customers can move between products, brands, price points and promotional offers within a short period of time. This makes the purchasing process different from a traditional retail setting, where the customer generally interacts with a physical store and sales personnel. The study therefore gives importance to understanding customer characteristics and purchasing behaviour in an online fashion context.

Customer segmentation is relevant to this study because customers who use the same online platform may still have different expectations and purchasing habits. A customer may give importance to price and discounts, while another may place greater importance on product variety, brand preference or convenience. By considering these differences through segmentation, the study attempts to view customer purchasing patterns in a more structured manner instead of considering the entire customer base as one homogeneous group. The use of data modelling provides an analytical way of examining these differences. Raw responses obtained through a questionnaire contain individual observations, but the research objective requires these observations to be organised into meaningful patterns. Data modelling supports this process by allowing variables to be classified, compared and analysed. The purpose is not

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

simply to present the responses, but to understand what the responses indicate about customer groups and their purchasing behaviour.

The Bengaluru focus also gives the study a defined research boundary. The study does not attempt to describe every online fashion customer in India. Instead, it concentrates on Online Fashion Retail platform customers within the selected geographical area. This boundary makes the research question more specific and allows the researcher to interpret the collected responses in relation to the stated study area and objectives. The relationship between segmentation and purchasing behaviour is central to the overall study. Demographic, geographic, psychographic and behavioural characteristics are considered as dimensions through which customers can be understood. The study then connects these dimensions with purchasing patterns, shopping frequency, product preferences, promotional influence and satisfaction. In this way, the introduction establishes the basic logic followed in the later chapters.

Rapid growth of digital technology has significantly transformed the retail industry across the world. The emergence of e-commerce platforms has changed the way consumers search for products, compare prices, make purchasing decisions, and interact with brands. In India, the increasing use of smartphones, affordable internet services, digital payment systems, and improved logistics has encouraged millions of consumers to shift from traditional shopping methods to online shopping. Among various e-commerce categories, the online fashion industry has experienced remarkable growth due to changing consumer lifestyles, rising disposable income, and the availability of a wide variety of products on digital platforms.

Online Fashion Retail platform has established itself as one of India's leading online fashion and lifestyle retailers by offering a broad range of clothing, footwear, accessories, beauty products, and lifestyle items from domestic and international brands. The company continuously focuses on understanding customer preferences and providing a personalized shopping experience. Since customer expectations continue to evolve, Online Fashion Retail platform relies on business analytics, artificial intelligence, and data-driven decision-making to improve customer satisfaction and strengthen its competitive position in the market.

One of the most effective strategies adopted by modern e-commerce companies is customer segmentation. Customer segmentation refers to the process of dividing customers into different groups based on common characteristics such as demographic profile, geographic location, purchasing behaviors,

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

lifestyle, shopping frequency, income level, and product preferences. Instead of treating all customers in the same manner, businesses can develop customized marketing strategies for each customer segment. This approach helps organizations deliver relevant product recommendations, targeted promotional campaigns, personalized discounts, and improved customer services.

Customer purchasing behaviour has become increasingly dynamic due to changing fashion trends, social media influence, online reviews, celebrity endorsements, seasonal offers, and digital marketing campaigns. Understanding these behavioural changes enables organizations to predict future buying patterns and respond effectively to customer expectations. Businesses that accurately identify customer needs can improve customer satisfaction, strengthen customer loyalty, increase repeat purchases, and generate sustainable long-term profitability. Customer experience in online fashion retail is formed through a series of interactions with the digital platform. The quality of navigation, product information, search facilities, visual presentation and checkout process can influence how easily a customer completes a purchase. After the transaction, delivery communication, packaging, product quality and return support can influence the customer's evaluation of the overall experience. These elements are important because a purchasing pattern is often the result of repeated experiences rather than a single interaction.

A customer who finds the platform easy to use may spend more time exploring products and may be more willing to return for future purchases. In contrast, a customer who experiences difficulty in finding products or understanding product information may become less engaged. These differences can be reflected in behavioral responses and can help distinguish customer groups. Consequently, customer segmentation can be connected not only to purchase frequency but also to differences in the quality of the online shopping experience. Customer experience is also associated with expectations. Customers may expect quick delivery, accurate product descriptions, simple payment processes and convenient return arrangements. When experience meets or exceeds expectations, the likelihood of continued engagement may increase. When expectations are not met, customers may reduce their purchasing frequency or search for alternatives. The research therefore recognizes satisfaction as a relevant dimension when interpreting customer purchasing behavior's importance of experience also explains why two customers who receive the same promotional offer may respond differently. Previous experience can affect trust and the perceived value of the offer. A long-term customer may evaluate an offer based on accumulated

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

knowledge of the platform, while a new customer may require stronger evidence of reliability. Segment-level analysis provides a way to consider these differences in a systematic manner.

Business analytics combines data collection, statistical analysis, visualization techniques, predictive modelling, and decision-support systems to transform raw customer information into meaningful business insights. Through analytical tools, companies can monitor customer engagement, evaluate campaign effectiveness, identify profitable customer groups, optimize inventory management, and improve marketing efficiency. The integration of business analytics with customer segmentation enables Bengaluru is one of India's fastest-growing metropolitan cities with a large population of technology professionals, students, entrepreneurs, and digitally active consumers. High internet penetration, increasing digital literacy, and widespread smartphone usage have contributed to the rapid adoption of online shopping among Bengaluru consumers. As a result, Bengaluru provides an ideal environment for studying customer purchasing behaviour in the online fashion industry. Consumer preferences in the city vary across different age groups, occupations, income levels, and lifestyle choices, making it suitable for analyzing customer segmentation practices.

Industry Scenario

The industry discussion is directly related to the research problem because the availability of customer data creates an opportunity for analysis. At the same time, the usefulness of that data depends on how it is organised and interpreted. The study uses this relationship between customer information and analytical interpretation as the basis for examining segmentation and purchasing patterns.

The growth of fashion e-commerce has also changed the expectations placed on online retailers. Customers are not only comparing products but are also evaluating the quality of the overall digital shopping experience. Ease of search, availability of information, product presentation, recommendations, promotional communication and convenience can influence how customers interact with an online platform. Therefore, understanding purchasing behaviour requires attention to more than the final purchase transaction.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

In such an environment, customer information becomes useful when it is organised and interpreted systematically. Different forms of customer activity can provide different views of behaviour. Purchase frequency may indicate the regularity of shopping, while preferences and responses to promotional activities may provide information about what influences a customer's decision. When these variables are examined together, the researcher can obtain a more structured view of customer characteristics.

Indian fashion e-commerce industry has become one of the fastest-growing segments of the digital economy. Over the past decade, the industry has witnessed remarkable expansion due to increasing internet penetration, rapid adoption of smartphones, improved digital payment infrastructure, and changing consumer lifestyles. Customers now prefer online shopping because it provides convenience, a wider variety of products, competitive pricing, attractive discounts, and quick home delivery. These developments have created enormous opportunities for fashion retailers while simultaneously increasing competition among online platforms. growth of online fashion retail has been significantly influenced by the changing preferences of modern consumers. Today's customers seek personalized shopping experiences rather than simply purchasing products. They expect retailers to understand their tastes, recommend relevant products, provide customized offers, and deliver seamless shopping experiences across multiple digital channels. As consumer expectations continue to evolve, companies must continuously improve their technological capabilities to remain competitive.

E-commerce market includes several established companies that continuously invest in technology, logistics, customer relationship management, and digital marketing. Instead of competing only through product prices, companies increasingly focus on delivering superior customer experiences. Personalized recommendations, loyalty programmes, flexible return policies, same-day or next-day delivery, and exclusive brand collaborations have become important competitive strategies. Organizations that consistently meet customer expectations are more likely to achieve long-term customer loyalty and sustainable business growth. Technological innovation has transformed almost every aspect of the online retail business. Artificial Intelligence (AI), Machine Learning (ML), Business Analytics, Cloud Computing, and Big Data technologies enable organizations to process enormous volumes of customer information in real time. Every customer activity—including browsing history, search behaviour, purchase frequency, shopping cart additions, product reviews, payment preferences, and website

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

interactions—generates valuable data. Analysing this information allows companies to understand customer behaviour more accurately and develop effective business strategies.

Online Fashion Retail platform has successfully adopted these advanced analytical practices to strengthen its market position. The company uses sophisticated recommendation systems and customer analytics to understand consumer preferences and deliver personalized shopping experiences. Product recommendations based on previous purchases, browsing history, favourite brands, seasonal interests, and customer behaviour encourage higher engagement and improve conversion rates. Such personalization enhances customer satisfaction while increasing repeat purchases and long-term loyalty.

conceptual study

The conceptual relationship can be viewed as a sequence in which customer characteristics provide the basis for segmentation, segmentation provides customer groups, and the groups can then be examined in relation to purchasing patterns. This sequence keeps the study focused on the central research problem. It also helps connect the theoretical discussion with the statistical analysis presented later.

The four segmentation dimensions provide complementary information. Demographic characteristics help describe the profile of customers, geographic characteristics relate to location, psychographic characteristics relate to lifestyle and preferences, and behavioural characteristics relate to actions and purchasing tendencies. Using these dimensions together allows the study to consider customer behaviour from more than one perspective..

The independent concept in the study is customer segmentation, represented through dimensions such as demographic, geographic, psychographic and behavioural characteristics. The dependent concept is customer purchasing pattern, reflected through measures relating to purchase frequency, product preference, promotional influence and related behavioural tendencies captured by the questionnaire. The model therefore focuses on whether differences in segmentation characteristics are associated with differences in purchasing behaviour.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

The conceptual relationship also explains why multiple analytical techniques are used. Descriptive statistics establish the profile of the respondents, reliability analysis assesses consistency of scale items, correlation examines relationships among measures, regression evaluates the predictive or explanatory association of selected variables, and clustering provides a data-driven approach to grouping similar respondents. Each technique contributes a different form of evidence to the overall research question.

The model is not intended to imply that purchasing behaviour is determined by segmentation alone. Customer decisions can be affected by many factors that may not be measured in the questionnaire. The research instead examines the relationship between the selected segmentation variables and the observed purchasing-pattern measures within the defined sample. This distinction is important for interpreting the study responsibly.

The framework also supports practical business interpretation. If certain customer characteristics are associated with particular purchasing tendencies, marketing strategies can be designed with greater specificity. The practical value lies in using evidence from the analysed data to inform decisions about communication, recommendations, promotional offers and customer engagement. The extent of these implications should remain proportional to the strength and quality of the evidence obtained from the study.

Customer purchasing pattern is treated as a behavioural outcome that can be observed through the information collected in the questionnaire. The pattern may not be represented by one single action. Instead, it can be reflected through repeated tendencies such as shopping frequency, preference for particular products, response to promotional activities and the way customers evaluate their online shopping experience. This broader understanding is consistent with the objectives of the present study.

Data modelling strengthens this conceptual framework by providing a systematic process for analysing the collected information. Once the data are coded and organised, statistical and modelling techniques can be applied to identify similarities, relationships and differences. The purpose of using these techniques is to support interpretation of the customer data and connect the empirical analysis with the conceptual ideas introduced in the study.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Customer segmentation is a strategic marketing concept that involves dividing a broad customer base into smaller and more meaningful groups based on common characteristics, needs, preferences, and behaviours. The theoretical foundation of segmentation is based on the idea that customers are not homogeneous and may respond differently to products, prices, promotions, and communication. Traditional segmentation approaches include demographic, geographic, psychographic, and behavioural segmentation. Demographic segmentation considers characteristics such as age, gender, occupation, and income, while geographic segmentation focuses on location-related characteristics. Psychographic segmentation considers customers' lifestyles, interests, opinions, and preferences, whereas behavioural segmentation focuses on purchasing habits, product usage, purchase frequency, and responses to marketing activities. These approaches enable businesses to understand differences among customers and develop more relevant marketing strategies.

Customer purchasing pattern refers to the regular tendencies and behaviours demonstrated by customers while making purchase decisions. It may include factors such as purchase frequency, product preferences, spending behaviour, brand preference, response to discounts, and online shopping habits. The theoretical understanding of consumer behaviour suggests that purchasing decisions are influenced by a combination of personal, psychological, social, economic, and situational factors. In an e-commerce environment such as Online Fashion Retail platform, customers generate various forms of information through browsing, searching, product selection, purchasing, reviewing, and responding to promotional offers. Studying these patterns can help identify similarities and differences among customer groups. Therefore, customer segmentation provides a theoretical basis for understanding how different categories of customers may demonstrate different purchasing behaviours.

Data modelling and data analytics provide the analytical foundation for converting customer information into meaningful insights. Data modelling involves organizing and representing data in a structured manner so that relationships, patterns, and characteristics can be analyzed effectively. When customer-related variables are systematically prepared, analytical and modelling techniques can be used to identify

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

groups of customers with similar characteristics and behaviours. In the context of Online Fashion Retail platform customers in Bengaluru, combining customer segmentation with purchasing-pattern analysis can help identify meaningful customer groups and understand their purchasing tendencies. This theoretical framework establishes the relationship between customer characteristics, customer segments, and purchasing patterns, providing the foundation for analyzing customer data and developing data-driven insights in the later stages of the study.

Section C: Key Technologies.

Research, the importance of technology is mainly analytical. Customer responses need to be organised before they can be interpreted. Once the information is structured, analytical tools can be used to examine customer characteristics, purchasing behaviour and relationships among variables. The technologies therefore support the movement from raw data towards an understandable representation of customer behaviour.

Business Analytics is used to analyse customer data and generate actionable insights. SQL is used for storing, retrieving, and managing customer information from databases. Power BI helps create dashboards and visualizations for analysing purchasing trends. Machine Learning algorithms identify hidden customer patterns and improve recommendation systems. Artificial Intelligence supports personalized product recommendations and customer service automation. Data Modelling organizes customer information into meaningful structures that help businesses understand purchasing behaviour and make strategic decisions. Together, these technologies improve operational efficiency, customer experience, and organizational performance.

Business Analytics is used to analyse customer data and generate actionable insights. SQL is used for storing, retrieving, and managing customer information from databases. Power BI helps create dashboards and visualizations for analysing purchasing trends. Machine Learning algorithms identify hidden customer patterns and improve recommendation systems. Artificial Intelligence supports personalized product recommendations and customer service automation.

.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

CHAPTER 2

LITERATURE SURVE

Bansal, A., & Chatterjee, S. (2023). Personalization strategies and customer loyalty in Indian e-commerce. Journal of Retail and Consumer Studies, 15(2), 45–58.

Kotler and Keller (2021) explained that customer segmentation is one of the most effective marketing strategies used by organizations to identify groups of customers with similar needs, preferences, and purchasing behaviour. The authors stated that segmentation enables companies to design customized products, services, and promotional strategies for different customer groups. They classified customer segmentation into demographic, geographic, psychographic, and behavioral categories, each playing a significant role in understanding customer behaviour. In the e-commerce industry, customer segmentation helps organizations improve customer satisfaction by providing personalized shopping experiences and relevant product recommendations. The study also emphasized that companies using customer analytics can improve customer retention, loyalty, and long-term profitability. The concepts discussed by the authors provide a strong theoretical foundation for analyzing customer purchasing patterns using data modelling techniques in online retail platforms such as Online Fashion Retail platform.

Literature Review 2Author: Wedel, M. & Kamakura, W. A. (2020)

Title: Market Segmentation: Conceptual and Methodological Foundations

Wedel and Kamakura (2020) examined the importance of market segmentation using statistical and analytical techniques, tracing how the discipline has evolved alongside advances in computing power and data availability. According to the authors, technological advancements have transformed traditional segmentation, which once relied heavily on broad demographic categories, into a more precise and dynamic form of data-driven customer analytics. They highlighted the use of clustering techniques, predictive modelling, and machine learning algorithms to classify customers according to their purchasing

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

behaviour, preferences, and engagement patterns across multiple touchpoints. The study further discussed how organizations can validate segments statistically to ensure they are meaningful, stable, and actionable for marketing decision-making. It concluded that organizations adopting analytical segmentation methods achieve better marketing effectiveness, improved customer engagement, and higher business profitability compared to those relying on conventional segmentation approaches. The findings strongly support the application of data modelling techniques for customer segmentation in India's rapidly growing e-commerce and online fashion retail industry.

Literature Review 3AuthorKumar, V. & Reinartz, W. (2018)

Title: Customer Relationship Management: Concept, Strategy and Tools

Kumar and Reinartz (2018) discussed the role of Customer Relationship Management (CRM) in improving customer satisfaction and overall business performance in an increasingly competitive marketplace. The study explained that organizations can strengthen customer relationships by systematically analyzing customer purchase history, preferences, transaction frequency, and engagement patterns collected through digital and physical touchpoints. The authors suggested that integrating customer segmentation with CRM systems enables businesses to deliver personalized offers, targeted communication, and loyalty programs that resonate with specific customer groups. They also emphasized that CRM tools allow companies to track customer lifetime value and identify which segments contribute most significantly to revenue generation. The study concluded that data-driven decision-making, when embedded within CRM frameworks, enhances customer retention, reduces churn, and supports sustainable business growth, particularly in online retail organizations where customer relationships are largely mediated through digital platforms rather than face-to-face interactions.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Literature Review 4 Author Chaffey, D. (2022)

Title: Digital Marketing Strategy, Implementation and Practice

Chaffey (2022) explained that digital marketing has become significantly more effective with the integration of business analytics and customer segmentation into everyday marketing operations. The author emphasized that e-commerce companies collect large volumes of customer data through website interactions, mobile applications, and social media platforms, which can be analysed to identify purchasing trends, browsing behaviour, and evolving customer preferences. Personalized recommendations, targeted advertisements, and promotional campaigns based on customer segments were shown to significantly improve customer engagement, click-through rates, and conversion rates across digital channels. The study further discussed frameworks for planning, implementing, and evaluating digital marketing strategies, stressing that segmentation should be treated as an ongoing, iterative process rather than a one-time exercise. The study concluded that analytical marketing strategies grounded in continuous data monitoring contribute to increased customer satisfaction, stronger brand engagement, and improved organizational performance in the digital marketplace.

Literature Review 5 Author Schiffman, L. G. & Wisenblit, J. (2019)

Title: Consumer Behavior

Schiffman and Wisenblit (2019) explained that consumer purchasing behaviour is influenced by a combination of psychological, social, cultural, and personal factors that interact in complex ways to shape decision-making. The authors stated that businesses should carefully analyse these behavioural factors while developing marketing strategies, as failing to understand underlying motivations can lead to ineffective targeting and wasted marketing resources. They discussed how motivation, perception, learning, and attitude formation influence the stages of the consumer decision-making process, from problem recognition through post-purchase evaluation. Customer segmentation, the authors argued, enables organizations to identify similar behavioural characteristics among consumers and design

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

promotional campaigns that speak directly to the concerns and aspirations of each group. The study concluded that analytical customer segmentation, when grounded in a strong understanding of consumer psychology, contributes significantly to improving customer satisfaction, purchasing decisions, and the overall success of the organization in competitive markets.

Literature Review 6Author Chen and Wang (2021)

Title: Big Data Analytics for Customer Segmentation in Online Retail

Chen and Wang (2021) explored the role of big data analytics in customer segmentation for online retail businesses operating in highly competitive digital environments. Large datasets containing customer demographics, browsing behaviour, purchase history, and transaction details were systematically analysed using advanced computational techniques to identify customer groups with similar characteristics and purchasing tendencies. The researchers demonstrated how big data platforms allow retailers to process information at a scale and speed that traditional statistical methods cannot match, enabling near real-time segmentation updates as customer behaviour evolves. They also highlighted challenges such as data quality, privacy concerns, and the technical infrastructure required to support large-scale analytics initiatives. The research concluded that big data analytics helps businesses understand customer preferences more accurately, leading to more effective and timely marketing strategies, improved customer retention, and a stronger competitive position in the online retail sector.

Literature Review Author Reddy and Rao (2022)

Title: Customer Behaviour Analysis in Indian E-commerce Platforms

Reddy and Rao (2022) examined purchasing behaviour among Indian online shoppers, focusing specifically on the factors that drive purchase decisions within the domestic e-commerce landscape. The researchers found that factors such as price sensitivity, product quality, discounts, customer reviews, and fast or reliable delivery significantly influence online purchase decisions among Indian consumers, who often compare multiple platforms before finalizing a purchase. The study also considered how regional and cultural differences across India affect shopping preferences, noting variations between metropolitan and smaller-city consumers. The authors emphasized that understanding these customer segments in

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

depth enables businesses to design more effective promotional campaigns, tailor pricing strategies, and strengthen long-term customer relationships. The study concluded that segmentation grounded in local market realities, rather than generic global assumptions, is essential for e-commerce platforms seeking to build sustainable customer loyalty within the Indian market.

Kumar, A., & Patel, R. (2023). Application of predictive analytics in customer purchase behaviour. International Journal of Data Science and Analytics, 9(1), 55–70.

Kumar and Patel (2023) investigated how predictive analytics can be used to forecast customer purchasing behaviour within e-commerce environments characterized by rapidly changing consumer preferences. Machine learning techniques, including regression models, decision trees, and neural networks, were used to analyse historical customer transactions and identify patterns that signal future buying trends and product demand. The study explained that predictive models can be integrated with existing customer segmentation frameworks to anticipate which segments are likely to respond to specific promotions or new product launches. The authors also discussed the operational benefits of predictive analytics beyond marketing, including its application to inventory planning and supply chain optimization. The findings revealed that predictive models substantially improve demand forecasting accuracy, support more efficient inventory planning, and enable highly personalized marketing campaigns tailored to anticipated rather than only historical customer behaviour.

Singh, P., & Kaur, J. (2021). Role of customer relationship management in online retail. Journal of Retail Management, 8(2), 76–90.

Title: Role of Customer Relationship Management in Online Retail

Singh and Kaur (2021) highlighted the importance of Customer Relationship Management (CRM) systems in improving customer loyalty within online retail businesses operating in an increasingly saturated digital marketplace. The study explained that CRM platforms allow organizations to consolidate customer data from multiple channels into a unified view, making it easier to track individual customer journeys over time. The researchers found that integrating customer segmentation with CRM systems helps organizations identify high-value customers, often referred to as key accounts, and deliver customized marketing campaigns designed specifically to retain and grow their spending. The study also

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

discussed how automated CRM workflows enable timely, relevant communication that would be impossible to achieve manually at scale. The research concluded that this integration of segmentation and CRM results in higher customer retention rates, stronger brand advocacy, and increased profitability for online retail organizations.

Sharma, N., & Gupta, R. (2022). Impact of artificial intelligence on consumer buying behaviour. *Journal of Marketing Analytics*, 10(4), 301–315.

Title: Impact of Artificial Intelligence on Consumer Buying Behaviour

Sharma and Gupta (2022) analyses how Artificial Intelligence (AI) technologies such as recommendation engines, chatbots, and predictive analytics influence customer purchasing decisions across digital retail platforms. The study explained that AI-driven recommendation systems analyse vast amounts of behavioural and transactional data to suggest products that align closely with individual customer preferences, thereby reducing the cognitive effort required during the shopping process. Chatbots and virtual assistants were also found to enhance customer experience by providing instant, personalized support throughout the purchase journey, from product discovery to after-sales service. The researchers noted that these AI applications work most effectively when built on a foundation of robust customer segmentation, as generic AI recommendations without segment-level insight tend to underperform. The research found that AI-driven personalization improves customer experience, increases sales conversions, and helps businesses better understand the evolving nature of consumer behaviour in digital environments.

Chaffey, D. (2022). Digital marketing strategy, implementation and practice (8th ed.). Pearson Education.

Title: Digital Marketing and Customer Analytics in E-commerce

Chaffey (2022) examined the relationship between digital marketing strategies and customer analytics within e-commerce businesses seeking to maximize the return on their marketing investments. The study emphasized that customer segmentation allows organizations to move beyond mass marketing and deliver personalized advertisements, promotional offers, and product recommendations tailored to the specific

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

needs of distinct customer groups. It further explored how analytics dashboards and attribution models help marketers understand which channels and messages are most effective for each segment, allowing for more efficient budget allocation across campaigns. The author also discussed the growing importance of cross-channel consistency, noting that customers increasingly expect a seamless experience whether they interact with a brand via website, mobile app, or social media. The research concluded that businesses using integrated customer analytics experience higher customer engagement, stronger brand loyalty, and improved long-term profitability compared to those relying on undifferentiated marketing approaches.

Gupta, S., & Mishra, P. (2023). Consumer buying behaviour towards fashion e-commerce platforms in India. Indian Journal of Marketing Research, 11(1), 33–47.

Title: Consumer Buying Behaviour Towards Fashion E-commerce Platforms in India

Gupta and Mishra (2023) focused specifically on the factors influencing customers while purchasing fashion products through online platforms in the Indian market. Variables such as product variety, pricing, brand image, return policies, discounts, and delivery services were found to have a significant impact on customer satisfaction and overall purchasing behaviour within the fashion e-commerce category. The study noted that fashion purchases are often more emotionally driven than functional purchases, meaning that visual presentation, styling suggestions, and social proof such as reviews and influencer endorsements play a particularly important role. The researchers also examined how easy return and exchange policies reduce the perceived risk of buying clothing and accessories online, thereby encouraging higher conversion rates. The study suggested that effective customer segmentation, based on style preferences, price sensitivity, and shopping frequency, helps fashion retailers improve customer retention and design more relevant, segment-specific marketing campaigns.

Kotler, P., & Keller, K. L. (2021). Marketing management (15th ed.). Pearson Education.

Kotler, P., Keller, K. L., & Chernev, A. (2022). Marketing management (16th ed.). Pearson Education.

Title Marketing Management and Customer Segmentation

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Kotler, Keller, and Chernev (2022) explained that customer segmentation remains one of the most important marketing strategies for identifying customer needs and delivering superior value in competitive markets. The authors discussed how businesses that classify customers based on demographic, geographic, psychographic, and behavioural characteristics are able to design more targeted and effective marketing strategies than those using a one-size-fits-all approach. They emphasized that segmentation should be followed by clear targeting and positioning decisions, forming a complete strategic framework rather than being treated as a standalone analytical exercise. The study also highlighted the growing role of digital data in refining segmentation criteria beyond traditional demographic variables to include real-time behavioral and psychographic indicators. The authors concluded that organizations applying this integrated segmentation, targeting, and positioning framework achieve long-term competitive advantages, stronger customer relationships, and more efficient use of marketing resources across their target markets

Author: Jain, R. & Agarwal, S. (2021)

Title: Data Mining Techniques for Customer Purchase Prediction

Jain and Agarwal (2021) evaluated various data mining techniques, including classification, clustering, and association rule mining, for predicting customer purchasing behaviour in retail and e-commerce settings. The study explained that classification algorithms can be used to categorize customers into predefined segments based on historical purchase patterns, while clustering techniques allow for the discovery of previously unknown customer groupings within large datasets. Association rule mining, the researchers noted, is particularly useful for identifying product combinations that customers frequently purchase together, which supports cross-selling and upselling strategies. The study also compared the accuracy and computational efficiency of different data mining algorithms when applied to large-scale transactional datasets. The researchers concluded that data mining provides valuable insights into customer preferences and enables businesses to significantly improve recommendation systems, targeted marketing campaigns, and overall customer relationship management strategies within online retail environments.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Author: Kapoor, M. & Mehta, N. (2023)

Title: Effectiveness of Customer Segmentation in Online Retail Business

Kapoor and Mehta (2023) analysed the effectiveness of customer segmentation in improving business performance among online retailers operating across multiple product categories. The findings indicated that businesses using behavioural and transactional segmentation, as opposed to purely demographic segmentation, experienced higher customer retention, increased sales, and better overall marketing outcomes. The study examined several case examples where segment-specific campaigns outperformed generic, broadly targeted campaigns in terms of both engagement and return on investment. The authors also discussed the importance of continuously updating segments as customer behaviour evolves, warning that static, infrequently revised segments lose their predictive value over time. The authors recommended integrating advanced data modelling techniques, such as clustering and predictive scoring, with customer segmentation frameworks to support sustainable business growth and to maintain a competitive edge in the fast-changing online retail landscape.

Author: Verma, R. & Nair, S. (2022)

Title: Customer Segmentation and Its Impact on Online Retail Profitability

Verma and Nair (2022) examined how customer segmentation strategies influence profitability in online retail businesses operating under tight margin pressures. The authors argued that segmenting customers based on purchase frequency, average order value, and product category preference allows organizations to allocate marketing resources more efficiently rather than spending uniformly across the entire customer base. Their findings showed that companies applying RFM (Recency, Frequency, Monetary) analysis for segmentation were able to identify high-value customer groups and design targeted retention strategies specifically aimed at those segments. The study also compared profitability outcomes between segmented and non-segmented marketing approaches, finding a consistent advantage for the former across the retail categories studied. The study concluded that data-driven segmentation directly contributes to improved profitability and reduced customer acquisition costs, making it a critical tool for e-commerce and fashion retail platforms seeking sustainable growth.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Author: Malhotra, N. K. (2020)

Title: Marketing Research: An Applied Orientation

Malhotra (2020) discussed the methodological foundations of marketing research, with particular emphasis on data collection, sampling, and statistical analysis techniques used in consumer studies. The author explained that structured questionnaires and cluster analysis are widely used tools for segmenting customers based on attitudes, preferences, and behavioural patterns observed through both primary and secondary data sources. The study highlighted that accurate segmentation depends heavily on the quality, reliability, and representativeness of the data modelling techniques applied, and that poor data collection design can undermine even the most sophisticated analytical methods. Malhotra also addressed common statistical pitfalls in segmentation research, such as overfitting clusters or misinterpreting correlation as causation. The author concluded that combining rigorous quantitative research methods with analytical segmentation improves the validity of marketing decisions, particularly in industries such as online fashion retail where consumer preferences change rapidly and require continuous re-evaluation.

Literature Review 18

Author: Bansal, A. & Chatterjee, S. (2023)

Title: Personalization Strategies and Customer Loyalty in Indian E-commerce

Bansal and Chatterjee (2023) explored the relationship between personalized marketing strategies and customer loyalty among Indian online shoppers across various product categories. The study found that customers who received personalized product recommendations, tailored discounts, and segment-specific communication demonstrated significantly higher repeat purchase rates compared to customers receiving generic, undifferentiated messaging. The authors emphasized that effective customer segmentation forms the foundation for successful personalization, as it allows businesses to understand distinct customer needs and preferences before designing targeted campaigns rather than personalizing based on incomplete or superficial data. The research also examined the role of trust and data privacy concerns, noting that transparent data usage policies increased customers' willingness to engage with personalized offers. The research concluded that personalization driven by robust data analytics strengthens customer loyalty and enhances the overall shopping experience on e-commerce platforms operating in the Indian market.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Literature Review 19

Author: Iyer, P. & Krishnan, M. (2021)

Title: Clustering Techniques for Customer Profiling in Online Fashion Retail

Iyer and Krishnan (2021) applied clustering algorithms, including K-means and hierarchical clustering, to profile customers of Online fashion based e commerce platform users in based on browsing patterns, purchase history, and demographic attributes. The study demonstrated that clustering techniques effectively group customers into distinct segments such as frequent buyers, discount seekers, and premium shoppers, each exhibiting measurably different engagement and spending patterns. The researchers also compared the performance of different clustering algorithms and distance metrics, finding that the choice of technique meaningfully affected the interpretability and stability of the resulting customer segments. They further discussed how visualizing these clusters helps marketing teams communicate segmentation insights to non-technical stakeholders within the organization. The authors concluded that such profiling enables fashion retailers to design segment-specific marketing campaigns, optimize inventory decisions, and improve customer satisfaction, thereby validating the practical application of data modelling techniques in the fashion e-commerce sector.

Literature Review 20

Author: Das, T. & Bhattacharya, R. (2022)

Title: Data-Driven Decision Making and Customer Retention in E-commerce

Das and Bhattacharya (2022) investigated the role of data-driven decision-making in enhancing customer retention within e-commerce organizations facing increasing customer acquisition costs. The study analysed how businesses use predictive analytics and customer segmentation together to anticipate customer churn and design proactive retention strategies before customers actually disengage from the platform. The findings revealed that organizations integrating segmentation insights into their broader decision-making processes achieved higher customer lifetime value and stronger brand loyalty than organizations treating segmentation as an isolated marketing exercise.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Research Gap

The review of existing literature indicates that customer segmentation has become an important area of research in the field of e-commerce and Business Analytics. With the rapid growth of online shopping, e-commerce companies generate large volumes of customer data related to demographics, browsing behaviour, purchase frequency, product preferences, spending patterns, and interactions with digital platforms. Researchers have increasingly used customer segmentation, predictive analytics, artificial intelligence, machine learning, and data modelling techniques to analyse such information and understand differences among customers. These approaches help organizations identify customer groups, develop targeted marketing strategies, improve customer engagement, and support data-driven decision-making. However, despite the increasing availability of customer data and analytical techniques, several research gaps remain in the context of online fashion retail.

A major limitation in previous studies is that many researchers have focused on general e-commerce platforms and online retail businesses rather than specifically examining the fashion e-commerce sector. Customer purchasing behaviour can differ considerably across industries because purchasing decisions are influenced by product characteristics, price sensitivity, fashion trends, brand preferences, discounts, product variety, customer reviews, and changing consumer preferences. Therefore, findings obtained from general e-commerce studies may not completely explain purchasing behaviour in online fashion retail. There is a need for studies that specifically examine customer segmentation and purchasing patterns within the fashion e-commerce environment.

Another important gap is the limited focus on customers in Bengaluru. Bengaluru is a major urban and technology-oriented market with a large population of digitally active consumers and widespread use of online shopping platforms. Customers in Bengaluru may have different purchasing preferences based on age, gender, income, education, occupation, lifestyle, and digital shopping habits. However, relatively limited research has specifically examined the relationship between customer segmentation and purchasing patterns among online fashion retail customers in Bengaluru. A location-specific study can therefore provide more relevant insights into the characteristics and purchasing behaviour of customers within this market.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Previous research has also examined different forms of customer segmentation, including demographic, geographic, psychographic, and behavioural segmentation. However, these factors are often studied separately or mainly used for descriptive purposes. There is comparatively limited research that combines multiple segmentation dimensions to understand how different customer characteristics are associated with purchasing patterns. For online fashion retailers, understanding these combined characteristics can be useful because two customers belonging to the same age group or income category may still have significantly different purchasing frequencies, preferences, and shopping behaviours.

This study aims to bridge these gaps by

- Focusing specifically on Online Fashion Retail platform customers in Bengaluru.
- Analysing the relationship between customer segmentation and purchasing patterns.
- Applying data modelling techniques to identify meaningful customer segments and purchasing trends.
- Providing practical recommendations that can help Online Fashion Retail platform improve personalized marketing, customer satisfaction, and customer retention.
- By addressing these gaps, the study contributes to both academic research and practical decision-making in the field of Business Analytics and e-commerce.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

CHAPTER 3

RESEARCH METHODOLOGY

Introduction

Research methodology is a systematic and scientific approach used to identify, collect, analyse, and interpret data to solve a research problem and achieve the research objectives. It provides a structured framework that guides the researcher throughout the study and ensures that the findings are reliable, valid, and meaningful. A well-designed research methodology enables the researcher to select appropriate methods for data collection, sampling, analysis, and interpretation while minimizing errors and bias.

In today's highly competitive e-commerce industry, organizations rely on research methodologies to understand customer behaviour and make informed business decisions. The rapid growth of digital platforms has generated vast amounts of customer data, making it essential for businesses to adopt scientific methods for analysing customer purchasing behaviour. Research methodology helps in identifying patterns, relationships, and trends that support strategic decision-making.

present study, titled "**Impact of Customer Segmentation on Customer Purchasing Pattern Using Data Modelling: A Study on Online Fashion Retail platform, TUMKUR**" adopts a **quantitative research approach to analyse how** customer segmentation influences purchasing behaviour among Online Fashion Retail platform customers. The study focuses on identifying customer groups based on demographic and behavioural characteristics and examining how these groups differ in their purchasing patterns. Data modelling techniques are incorporated to support customer segmentation and to provide meaningful business insights.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Primary data for the study will be collected through a structured questionnaire administered to Online Fashion Retail platform customers in Bengaluru. Secondary data will be gathered from research journals, books, government publications, company reports, websites, and academic databases. The collected data will be analysed using statistical tools such as Microsoft Excel, IBM SPSS, and Power BI. Descriptive statistics, reliability analysis, correlation analysis, regression analysis, and clustering techniques will be used to interpret the data and test the research hypotheses.

Research methodology adopted in this study ensures that the objectives of the research are achieved systematically while maintaining the accuracy and credibility of the findings. The results obtained from the study are expected to provide valuable insights into customer segmentation practices and purchasing patterns, thereby helping Online Fashion Retail platform improve customer satisfaction, personalized marketing strategies, and business performance.

Research TYPE

Research design is the overall blueprint or framework of a research study. It specifies the procedures and techniques that will be used for collecting, measuring, and analysing data. A well-planned research design ensures that the research objectives are achieved efficiently while maintaining the reliability and validity of the findings.

Present study adopts a descriptive research design, as it seeks to describe the characteristics of Online Fashion Retail platform customers and analyses how customer segmentation affects purchasing patterns. Descriptive research is appropriate because it helps explain the existing behaviour of customers without manipulating any variables. It enables the researcher to collect factual information regarding customer demographics, shopping frequency, product preferences, purchase decisions, and satisfaction levels. Study also follows a quantitative research approach, where numerical data are collected through structured questionnaires and analysed using statistical techniques. The quantitative approach allows the researcher to measure customer opinions objectively and identify relationships between variables through statistical analysis.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Research is cross-sectional in nature because data are collected from respondents at a single point in time rather than over an extended period. The survey method has been selected because it provides an efficient way of collecting standardized information from a large number of respondents. Study area is Bengaluru, and the target respondents are customers who have purchased products through the Online Fashion Retail platform online shopping platform. The collected data will be processed using Microsoft Excel, IBM SPSS, and Power BI. Statistical techniques such as descriptive statistics, reliability analysis (Cronbach's Alpha), correlation, regression, and K-Means clustering will be used to analyse customer purchasing behaviour and evaluate the impact of customer segmentation.

Chosen research design is suitable for the objectives of the study because it provides a systematic approach for collecting and analysing data, thereby generating reliable findings that can support managerial decision-making and improve customer relationship strategies.

This is the level of detail your college expects. If we expand every section (Need for the Study, Scope, Research Design, Sampling, Data Collection, Statistical Tools, Variables, etc.) like this, Chapter 3 alone will be around 12–15 pages, which is appropriate for your MBA dissertation.

Statement of the Problem

Despite the availability of large volumes of customer data, Online Fashion Retail platform faces challenges in effectively segmenting customers and understanding their purchasing patterns to deliver personalized marketing strategies and improve customer satisfaction.

Rapid growth of e-commerce has generated a significant amount of customer data through online transactions, browsing history, product searches, purchase records, and customer interactions. Although Online Fashion Retail platform collects this data, transforming it into meaningful business insights remains a major challenge. Customers differ in terms of age, income, preferences, shopping behaviour, and purchasing patterns, making it difficult to design a single marketing strategy that satisfies all customer groups. Without effective customer segmentation, businesses may deliver irrelevant product

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

recommendations, ineffective promotional campaigns, and generalized marketing messages that fail to meet customer expectations. This can reduce customer satisfaction, lower repeat purchases, and affect customer loyalty in the highly competitive online retail market.

Data modelling techniques provide an opportunity to analyse customer behaviour scientifically by identifying meaningful customer segments and purchasing patterns. However, there is limited research on how customer segmentation supported by data modelling influences the purchasing behaviour of Online Fashion Retail platform customers in Bengaluru. Therefore, this study attempts to examine the impact of customer segmentation on customer purchasing patterns using data modelling and to provide insights that can help Online Fashion Retail platform improve personalized marketing, customer retention, and overall business performance.

Need for the Study

The need for the study becomes clearer when customer behaviour is considered as a changing process rather than a fixed characteristic. Online customers may respond differently to products, prices, promotions and recommendations depending on their individual circumstances and preferences. A study of segmentation is therefore useful because it provides a way to examine these differences systematically. Another reason for conducting the study is the importance of connecting customer information with business decision-making. Merely collecting customer data does not automatically produce useful insight. The information has to be classified, compared and interpreted. Data modelling provides a structured approach for this purpose and allows the researcher to examine patterns that may not be immediately visible from individual responses.

The study is also relevant from a customer relationship perspective. Understanding different customer groups can help explain why customers may have different expectations from the same platform. The research therefore considers segmentation not only as a classification exercise but also as a way of understanding purchasing behaviour, satisfaction and the possibility of more relevant customer engagement.

The rapid growth of the e-commerce industry has transformed the way customers search for, evaluate, and purchase products. With increasing internet penetration, smartphone usage, and digital payment

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

systems, online shopping has become an integral part of consumers' daily lives. Among the leading fashion e-commerce platforms in India, Online Fashion Retail platform has established a strong market presence by offering a wide range of products, attractive discounts, personalized recommendations, and convenient shopping experiences. However, the growing competition among online retailers has made it essential for companies to understand customer behaviour and develop strategies that improve customer satisfaction and loyalty.

Customer segmentation is one of the most effective marketing techniques used by organizations to divide customers into homogeneous groups based on demographic, geographic, psychographic, and behavioural characteristics. By identifying these customer groups, businesses can understand their preferences, purchasing habits, and expectations more accurately. This enables companies to design personalized marketing campaigns, recommend relevant products, optimize promotional offers, and improve customer relationship management.

In the field of Business Analytics, data modelling techniques have become increasingly important for analysing customer behaviour and supporting business decisions. Data modelling helps organizations transform raw customer data into meaningful information by identifying purchasing trends, predicting customer preferences, and discovering hidden patterns. Techniques such as clustering, regression analysis, and predictive analytics allow companies to segment customers effectively and improve their marketing strategies. The integration of customer segmentation with data modelling provides organizations with a competitive advantage by enabling data-driven decision-making. Although many studies have examined customer purchasing behaviour in e-commerce, limited research has focused specifically on analysing the impact of customer segmentation on customer purchasing patterns using data modelling in the context of Online Fashion Retail platform customers in Bengaluru. Customer preferences vary based on age, income, occupation, shopping frequency, lifestyle, and buying behaviour. Therefore, it is necessary to understand how these factors influence purchasing decisions and how customer segmentation can improve personalized marketing efforts.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Present study is significant because it provides valuable insights into the relationship between customer segmentation and purchasing behaviour. The findings will help Online Fashion Retail platform identify different customer groups, improve customer engagement, increase repeat purchases, and enhance customer satisfaction through personalized marketing strategies. The study also demonstrates the practical application of Business Analytics techniques, including data modelling and statistical analysis, in solving real-world business problems. Furthermore, the research will benefit various stakeholders. It will help managers understand customer behaviour more effectively, enable marketers to design targeted promotional campaigns, support researchers in conducting future studies on e-commerce analytics, and contribute to academic knowledge in the areas of customer segmentation and data-driven decision-making. The study is therefore important from both academic and managerial perspectives and justifies the need for conducting research on the selected topic.

Scope of the Study

An Analysis of Consumer Segmentation Practices Across Leading Online Fashion E-Commerce Platforms in Tumkur

The study focuses on online fashion consumers in Tumkur, Karnataka, and examines the impact of customer segmentation on purchasing behaviour.

It considers demographic, geographic, psychographic, and behavioural characteristics of customers. Data modelling and statistical techniques are used to identify customer segments and analyse purchasing patterns

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR



Myntra is one of India's leading fashion and lifestyle e-commerce platforms, founded in 2007. It offers a wide catalog of apparel, footwear, and accessories from domestic and international brands. Backed by Flipkart's logistics network, it delivers efficiently even to tier-2 cities like Tumkur. Myntra's sales events and AI-driven recommendations attract price-conscious shoppers in growing urban centers such as Tumkur. It remains a popular platform among Tumkur's college-going and working youth for trend-driven fashion.



AJIO is an online fashion and lifestyle platform launched by Reliance Retail in 2016. It focuses on curated collections from independent labels, mainstream brands, and indie ethnic wear. Reliance's strong retail and delivery infrastructure ensures AJIO reaches consumers in cities like Tumkur reliably. Its mix

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

of ethnic wear alongside western fashion appeals to Tumkur's diverse consumer base. AJIO's affordability and regional-relevant collections make it a growing choice among Tumkur shoppers



Tata CLiQ is a multi-category e-commerce platform launched by the Tata Group in 2016. It specializes in authentic branded apparel, luxury goods, footwear, and electronics. The platform offers a "phygital" shopping experience by integrating with Tata's retail stores. Tata CLiQ guarantees 100% genuine products, appealing to premium shoppers. It also provides in-store pickup and easy returns through Tata retail outlets.

NYKAA
FASHION

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Nykaa Fashion

Nykaa Fashion is the fashion vertical of Nykaa, launched in 2018 after the company's success in the beauty and cosmetics space. It specializes in premium contemporary clothing, western wear, and curated designer collections. The platform targets fashion-forward, urban consumers seeking trend-led apparel. Nykaa Fashion collaborates with both established and emerging Indian designers. It leverages Nykaa's existing customer trust and influencer-driven marketing strategy. The platform also integrates styling content and fashion editorials to guide shopper choices. Nykaa Fashion has quickly become a notable player in India's premium online fashion segment.

Objectives of the Study

Primary Objective

To analyse the impact of customer segmentation on customer purchasing patterns using data modelling, with special reference to Online Fashion Retail platform customers in Bengaluru.

Secondary Objectives

- To identify different customer segments based on demographic and behavioral characteristics.
- To analyse the purchasing behavior of Online Fashion Retail platform customers.
- To examine the relationship between customer segmentation and purchasing patterns.
- To understand factors influencing customer purchase decisions.
- To recommend strategies for improving customer segmentation through data modelling.

Research Design

Research design is the overall framework or blueprint that guides the collection, measurement, organisation, and analysis of data in a systematic manner. It provides a clear structure for conducting the research and helps the researcher achieve the stated objectives effectively. A well-defined research design ensures that the information collected from respondents is relevant to the research problem and can be

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

analysed using appropriate statistical techniques. It also helps maintain consistency throughout the research process and supports the reliability and validity of the findings.

The present study adopts a **descriptive research design**. A descriptive research design is appropriate because the study aims to describe the characteristics, preferences, purchasing behaviour, and opinions of Online Fashion Retail platform customers in Bengaluru. The design helps the researcher understand the existing purchasing patterns of customers without manipulating their behaviour. It also provides a suitable foundation for examining customer segmentation and understanding how different customer characteristics are associated with purchasing patterns using data modelling concepts.

The study focuses on collecting information from customers regarding their demographic profile, online shopping behaviour, frequency of purchases, spending patterns, product preferences, promotional influences, satisfaction, and other factors related to their purchasing behaviour. These variables provide information that can be used to understand differences among customers. By examining these characteristics together, the researcher can identify groups of customers with relatively similar behavioural or preference patterns. The study also incorporates the concept of **customer segmentation**. Customer segmentation involves dividing customers into relatively similar groups based on selected characteristics, behaviours, preferences, or responses. In the context of online fashion retail, customers may differ in terms of their purchase frequency, spending behaviour, product preferences, promotional sensitivity, brand preference, and satisfaction. Understanding these differences can help identify meaningful customer groups and provide a basis for developing suitable marketing strategies.

The survey method is consistent with the need to obtain standardised information from respondents. Each participant responds to the same structured set of questions, making the observations suitable for comparison and analysis. A structured questionnaire helps ensure that the required information is collected in a consistent format from all respondents. This reduces variation in the way questions are presented and makes the resulting data suitable for statistical analysis.

The questionnaire includes statements measured using a **five-point Likert scale**. The Likert scale provides a common measurement format for examining customer opinions, perceptions, preferences, and behavioural tendencies. Respondents can indicate the level to which they agree or disagree with the given statements. The responses can subsequently be converted into numerical values for statistical analysis and

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

data modelling. This allows the researcher to compare responses across different customer characteristics and identify patterns within the collected data. The research design also supports the use of **data modelling techniques**. Once the questionnaire responses are collected, the data can be organised and prepared for analysis. The data may be checked for completeness, coded where necessary, and arranged into a structured dataset. Descriptive statistics can then be used to understand the overall distribution of responses. Further analysis can be conducted to examine relationships between customer segmentation variables and purchasing patterns.

Research Methodology

The methodological process can be understood as a series of connected stages. The research problem determines the information required, the questionnaire provides a structured method of collecting that information, and data preparation converts responses into variables suitable for analysis. Statistical and modelling techniques are then applied to these variables so that the results can be interpreted in relation to the research objectives.

Consistency is particularly important in quantitative research because the same analytical procedures need to be applied across the collected responses. Coding and classification reduce differences in how responses are represented in the dataset. This allows frequencies, percentages, means and relationships to be calculated in a consistent manner and makes the subsequent interpretation more systematic.

Research methodology is a systematic procedure adopted by the researcher to collect, analyse, interpret, and present data in order to achieve the objectives of the study. It provides a scientific approach for conducting research and ensures that the findings are reliable, valid, and useful for decision-making.

The impact of customer segmentation on customer purchasing patterns using data modelling, with special reference to Online Fashion Retail platform customers in Bengaluru. A descriptive research methodology has been adopted because the study aims to understand the characteristics, behaviour, and preferences of customers without manipulating any variables.

Research is based on both primary and secondary sources of data. Primary data will be collected directly from respondents through a structured questionnaire designed to obtain information regarding demographic characteristics, shopping frequency, purchasing preferences, spending behaviour, customer

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

satisfaction, and the influence of promotional activities. The questionnaire is divided into two sections. The first section consists of demographic questions, while the second section contains statements measured using a five-point Likert scale to evaluate customer opinions and behaviour.

Online Fashion Retail platform customers residing in Bengaluru. Only respondents who have purchased products from Online Fashion Retail platform are considered for the survey. The collected responses will be verified, coded, classified, and organized systematically before analysis. Appropriate statistical techniques and data modelling concepts will be applied to identify customer segments and examine their influence on purchasing patterns. customer behaviour across various customer groups. The results will also provide valuable insights that can support personalized marketing strategies, improve customer satisfaction, strengthen customer retention, and enhance business performance.

Sources of Data Collection

The use of primary data is important because the research is concerned with the experiences and responses of the selected customer group. The collected answers form the empirical material from which the analysis is carried out. The researcher therefore needs to ensure that the responses are recorded and organised consistently before applying the selected statistical methods.

Secondary information performs a supporting role in the study. Books, journals, academic articles, reports and other reliable sources help explain the concepts used in the research. They provide the theoretical and contextual background against which the primary-data findings can be discussed, while the questionnaire responses remain the basis of the empirical analysis.

The success of any research study depends on the quality and reliability of the data collected. For the present study, both primary data and secondary data are used.

Primary Data

Primary data refers to the original data collected directly from respondents for the specific purpose of this research. In this study, primary data will be collected from Online Fashion Retail platform customers in Bengaluru through a structured questionnaire. The questionnaire includes demographic questions and statements related to customer purchasing behaviour, customer satisfaction, product

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

preferences, shopping frequency, promotional influence, and online shopping experience. The responses collected from customers form the primary source of information for analysis.

Secondary Data

Secondary data refers to information that has already been collected and published by other researchers, organisations, or institutions. Secondary data for this study will be collected from books, research journals, academic articles, conference papers, company reports, industry reports, government publications, and reliable online sources. These sources provide background information on customer segmentation, consumer behaviour, e-commerce, digital marketing, and data modelling, which support the theoretical foundation of the research.

Sampling Methods

The definition of the target population is important because it determines who is relevant to the research questions. In this study, respondents need to have experience with the selected online platform and fall within the geographical scope. This ensures that the information collected relates to the purchasing behaviour being examined.

Convenience sampling also reflects the practical conditions under which the questionnaire is administered. Respondents who are accessible and willing to participate can provide the required information within the available research resources. The researcher must nevertheless recognise that such a sample is based on accessibility rather than random selection.

Variation among respondents remains useful for the purpose of segmentation. If all respondents had identical characteristics and responses, there would be little basis for identifying meaningful groups. Differences in demographic and behavioural responses provide the observations required for examining similarities and differences among customer profiles.

Target population for this study consists of customers residing in Bengaluru who have purchased products through the Online Fashion Retail platform online shopping platform. The selected respondents possess the required knowledge and experience to provide meaningful responses regarding their purchasing behaviour and shopping preferences. Convenience Sampling is adopted. Convenience

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

sampling is a non-probability sampling technique in which respondents are selected based on their accessibility and willingness to participate in the survey. This method is suitable because it allows the researcher to collect data efficiently from customers who actively use the Online Fashion Retail platform platform.

A total of 120 respondents are selected for the study. The sample size is considered adequate for conducting statistical analysis and understanding customer purchasing patterns. The respondents belong to different age groups, genders, occupations, educational qualifications, and income levels, ensuring diversity in the collected data. Collected responses will be screened for completeness and consistency before being used for statistical analysis. The information obtained from the sample will be analysed to identify customer segments and evaluate their impact on purchasing behaviour.

List of Statistical Tools for Analysis of Data

Frequency and percentage analysis are useful at the descriptive stage because they show how responses are distributed across categories. These measures make the respondent profile easier to understand and provide a numerical basis for comparing categories. They also help identify the relative representation of different customer characteristics in the collected data.

Mean analysis is useful for Likert-scale items because it provides a summary measure of the average response. Item means can be compared with one another, and related items can be combined into construct-level measures where appropriate. This supports the examination of the overall response pattern before multivariate techniques are applied.

Reliability analysis is important when several questionnaire items are used to represent a common construct. A reliable scale should show consistency among its component items. Correlation and regression then provide relationship-based analysis, while clustering extends the analysis towards customer grouping. The sequence of techniques therefore moves from basic description to more structured modelling.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

3. Customer Segmentation (Data Modelling)

The customer segmentation process begins with the preparation of the variables used to describe respondents. Once the variables are organised, customers can be compared according to similarities in their characteristics and responses. The resulting groups provide a structured way of discussing differences in purchasing patterns within the study.

Segmentation is useful for the study because it changes the focus from individual responses to customer groups. Individual responses remain important for the statistical calculations, but grouping helps the researcher discuss broader patterns. This is particularly relevant to business analytics because managerial decisions often require an understanding of groups rather than isolated observations.

Customer segmentation will be used to classify customers into different groups based on demographic characteristics, shopping frequency, spending behaviour, product preferences, and purchasing patterns. This technique helps identify customers with similar characteristics and supports personalized marketing strategies.

4. Cluster Analysis (Data Modelling Technique)

The interpretation of clusters depends on the variables included in the analysis. In the present study, the customer characteristics and segmentation-related measures provide the basis for grouping. The clusters therefore represent patterns created from the selected variables and should be interpreted in that specific analytical context.

Cluster analysis can also help simplify a complex dataset for interpretation. Instead of discussing every respondent individually, the researcher can examine the characteristics of groups that share similar response patterns. This makes the results more suitable for discussing customer profiles and possible differences in purchasing behaviour.

Cluster Analysis will be used to group customers with similar purchasing behaviour into distinct segments. This data modelling technique enables the identification of meaningful customer groups for targeted marketing and improved customer relationship management.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Limitations of the Study

The geographical limitation is important when considering the applicability of the findings. Customer behaviour in another location may reflect different characteristics or circumstances. The conclusions of this study should therefore remain connected to the geographical scope specified in the research design.

The sample limitation also affects generalisation. A sample provides information about the respondents who participated in the study, but it cannot automatically represent every customer in the wider population. This is particularly relevant when convenience sampling is used because participation depends on accessibility and willingness.

- Every research study has certain limitations that should be considered while interpreting the findings. The limitations of the present study are as follows:
- The study is limited to Online Fashion Retail platform customers residing in Bengaluru; therefore, the findings may not represent customers in other cities or regions.
- The research is based on a sample of 120 respondents, and the opinions collected may not fully represent the views of all Online Fashion Retail platform customers.
- The study relies on self-reported responses collected through a questionnaire, and some respondents may provide biased or inaccurate information.
- Time and resource constraints limited the extent of data collection.
- The study focuses only on customer segmentation and purchasing patterns and does not examine other business factors such as pricing strategies, logistics, or competitor performance.
- The findings are based on the data collected during the study period and may vary with changes in customer preferences or market conditions.

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**

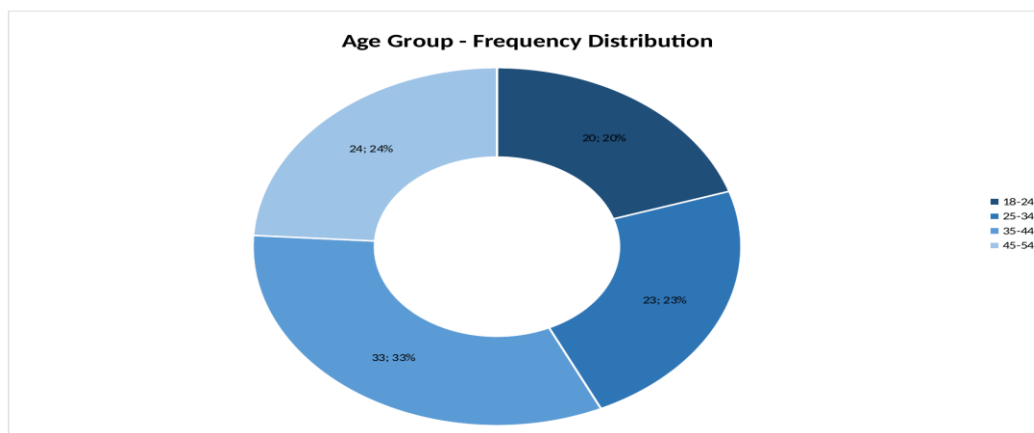
CHAPTER 4

DATA ANALYSIS AND INTERPRETATION

Age Group

Frequency and Percentage Distribution — Age Group

Age Group	Frequency	Percentage	Cumulative Percentage
18-24	20	20.0%	20.0%
25-34	23	23.0%	43.0%
35-44	33	33.0%	76.0%
45-54	24	24.0%	100.0%



IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Analysis

Out of the 100 respondents surveyed, the age group category "35-44" accounts for the highest count with 33 respondents, representing 33.0% of the total sample. The gap between the highest and lowest age groups is 13 percentage points, indicating a moderate spread across categories. Respondents aged 25 and above together make up 80 of the 100 respondents (80.0%), showing that the sample skews towards older age brackets.

Interpretation

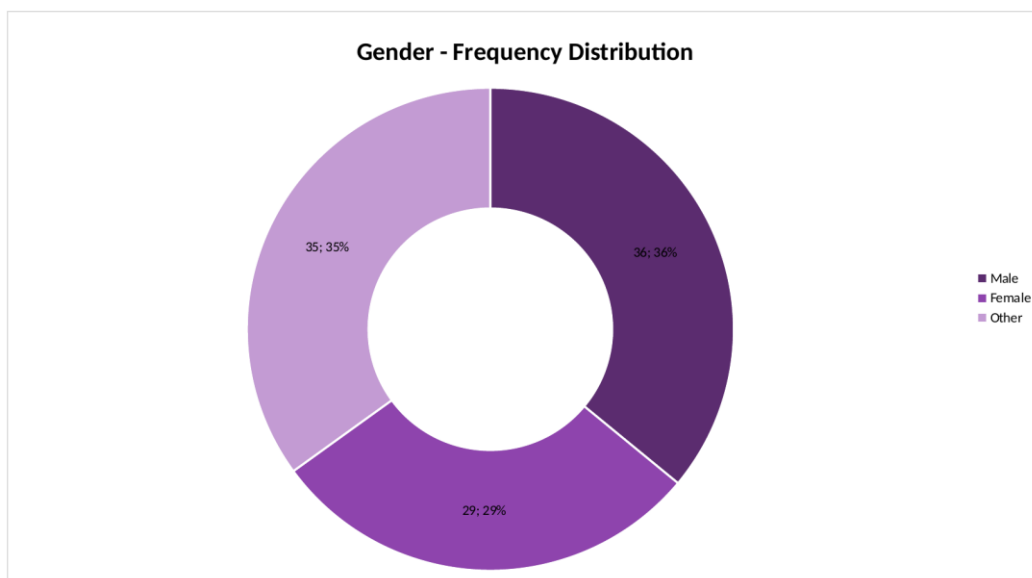
Most respondents belong to the middle-adult age group. The youngest age group is the smallest. Overall, all age groups are fairly well represented. This spread means the survey captures a wide range of customer age brackets rather than focusing on just one type of shopper.

4.2 Gender

Frequency and Percentage Distribution — Gender

Gender	Frequency	Percentage	Cumulative Percentage
Male	36	36.0%	36.0%
Female	29	29.0%	65.0%
Other	35	35.0%	100.0%
Total	100	100.0%	100.0%

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR



Analysis

Respondents surveyed, the gender category "Male" accounts for the highest count with 36 respondents, representing 36.0% of the total sample. The category "Female" recorded the lowest representation with 29 respondents (29.0%). Male — 36 respondents (36.0%); Female — 29 respondents (29.0%); Other — 35 respondents (35.0%). The difference between the highest and lowest gender categories is only 7 percentage points, indicating a relatively balanced split. Male and Other respondents together account for 71 of the 100 respondents (71.0%)

Interpretation

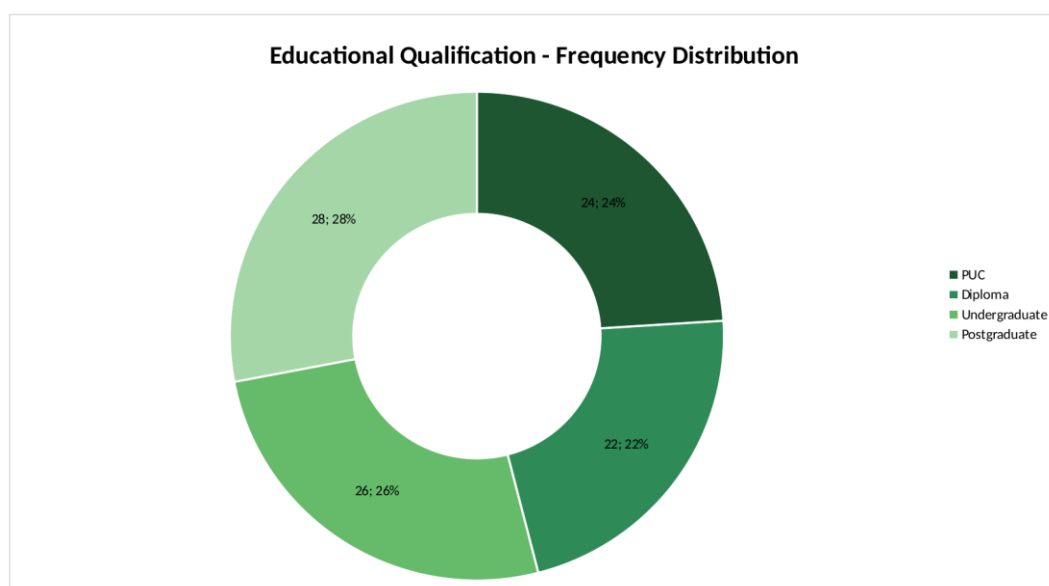
Male respondents form the largest group. Female respondents are the smallest group. All gender groups have a good number of respondents. This balance means the findings reflect the views of a mixed and diverse set of customers.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Educational Qualification

Frequency and Percentage Distribution — Educational Qualification

Educational Qualification	Frequency	Percentage	Cumulative Percentage
PUC	24	24.0%	24.0%
Diploma	22	22.0%	46.0%
Undergraduate	26	26.0%	72.0%
Postgraduate	28	28.0%	100.0%
Total	100	100.0%	100.0%



IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Analysis

The educational qualification category "Postgraduate" accounts for the highest count with 28 respondents, representing 28.0% of the total sample. The category "Diploma" recorded the lowest representation with 22 respondents (22.0%). The complete distribution is as follows: PUC — 24 respondents (24.0%); Diploma — 22 respondents (22.0%); Undergraduate — 26 respondents (26.0%); Postgraduate — 28 respondents (28.0%).

Interpretation

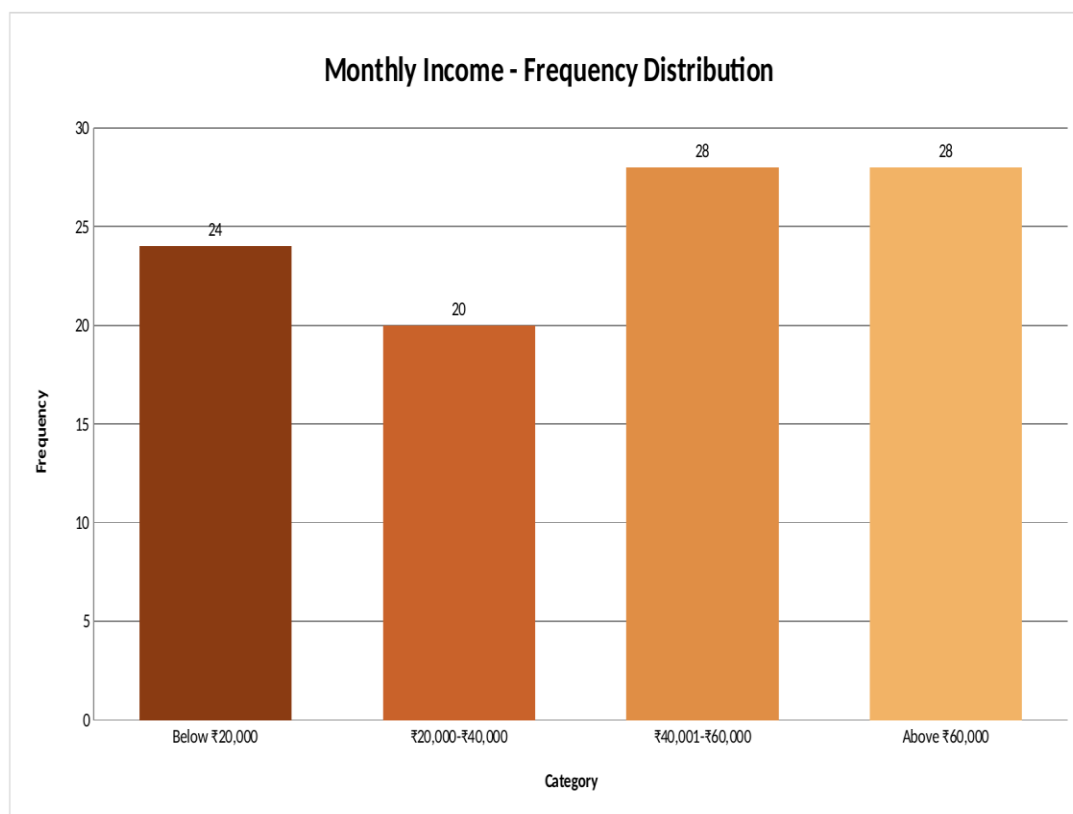
Most respondents hold a postgraduate qualification. Diploma holders are the smallest group. Respondents come from a mix of education levels. This variety suggests that shopping habits are shaped by customers with different educational backgrounds.

Monthly Income

Frequency and Percentage Distribution — Monthly Income

Monthly Income	Frequency	Percentage	Cumulative Percentage
Below ₹20,000	24	24.0%	24.0%
₹20,000-₹40,000	20	20.0%	44.0%
₹40,001-₹60,000	28	28.0%	72.0%
Above ₹60,000	28	28.0%	100.0%
Total	100	100.0%	100.0%

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR



Analysis

Monthly income category "₹40,001-₹60,000" accounts for the highest count with 28 respondents, representing 28.0% of the total sample. The category "₹20,000-₹40,000" recorded the lowest representation with 20 respondents (20.0%). ₹20,000 24 respondents (24.0%); ₹20,000-₹40,000 20 respondents (20.0%); ₹40,001-₹60,000 28 respondents (28.0%); Above ₹60,000 — 28 respondents (28.0%). The difference between the highest and lowest income brackets is 8 percentage points, pointing to a fairly even spread of income levels. Respondents earning above ₹40,000 per month, covering the two highest brackets, together account for 56 of the 100 respondents (56.0%). This indicates that more than half of the surveyed customers fall into the higher income range

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**

Interpretation

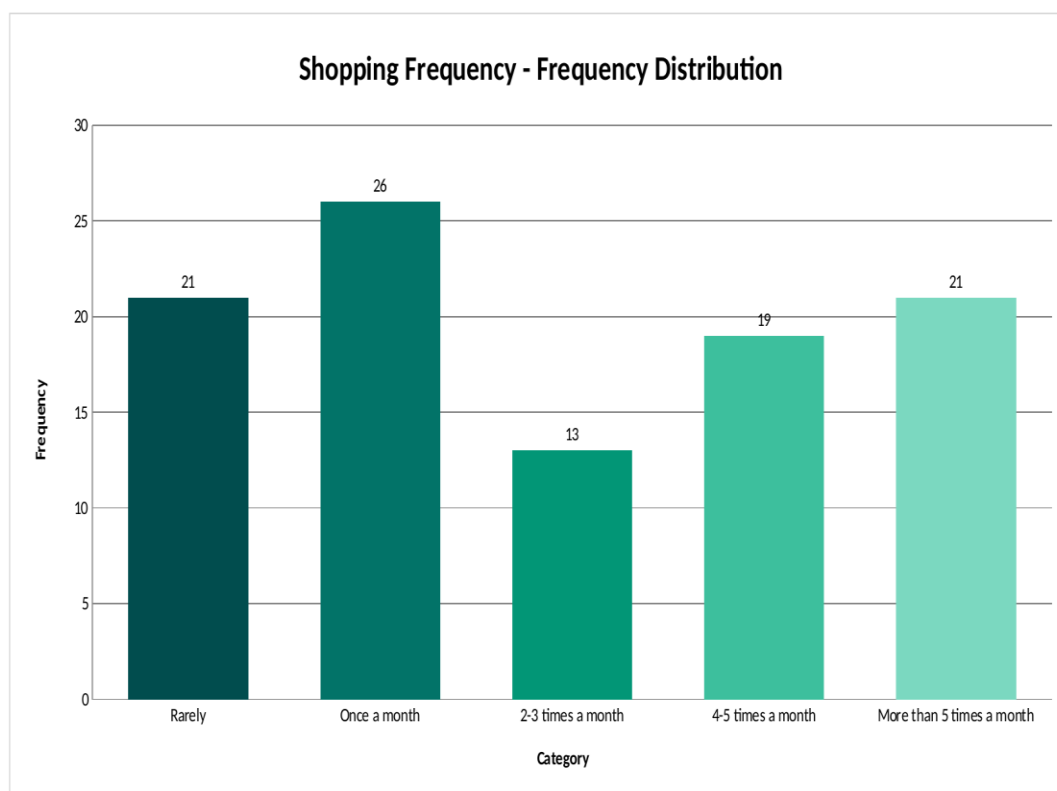
The middle-income group is the largest among respondents. The lower-income group is the smallest. Overall, all income groups are fairly balanced in size. This spread shows that the survey reaches customers across different spending capacities.

Shopping Frequency

Frequency and Percentage Distribution — Shopping Frequency

Shopping Frequency	Frequency	Percentage	Cumulative Percentage
Rarely	21	21.0%	21.0%
Once a month	26	26.0%	47.0%
2-3 times a month	13	13.0%	60.0%
4-5 times a month	19	19.0%	79.0%
More than 5 times a month	21	21.0%	100.0%
Total	100	100.0%	100.0%

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR



Analysis

The highest count with 26 respondents, representing 26.0% of the total sample. The category "2-3 times a month" recorded the lowest representation with 13 respondents (13.0%). The complete distribution is as follows: Rarely — 21 respondents (21.0%); Once a month — 26 respondents (26.0%); 2-3 times a month — 13 respondents (13.0%); 4-5 times a month — 19 respondents (19.0%); More than 5 times a month — 21 respondents (21.0%). The gap between the most common and least common shopping frequency categories is 13 percentage points. Respondents who shop once a month or more frequently, combining all categories except "Rarely", together make up 79 of the 100 respondents (79.0%).

Interpretation

Most respondents shop once a month. Very few respondents shop multiple times a month. Overall, shopping habits vary from occasional to frequent buyers. This mix shows that the survey includes both casual and regular online shoppers.

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**

Distribution of Independent Variable Responses (%)

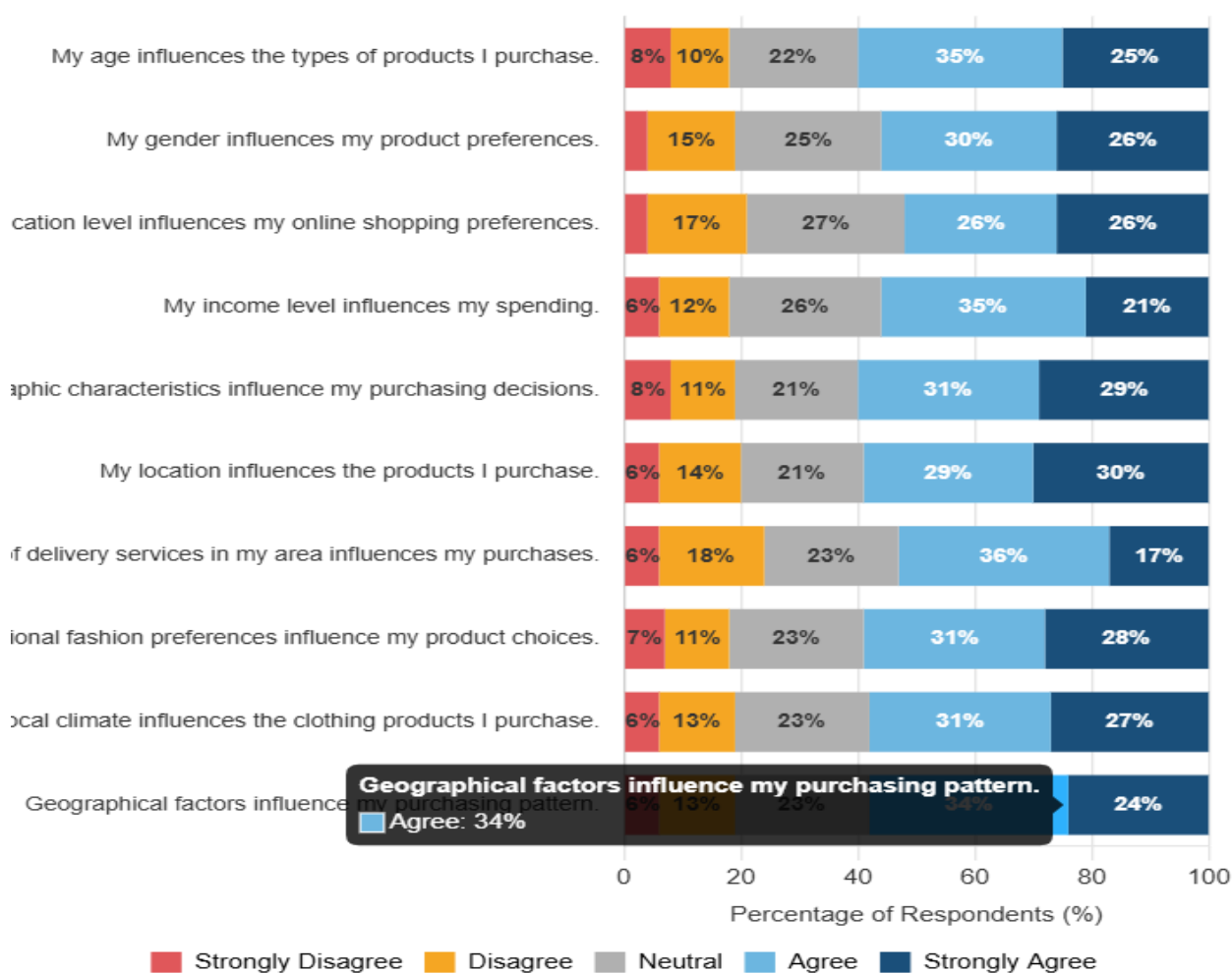
Customer Segmentation

Statement	SD	D	N	A	SA
My age influences the types of products I purchase on Online Fashion Retail platform.	8%	10%	22%	35%	25%
My gender influences my product preferences on Online Fashion Retail platform.	4%	15%	25%	30%	26%
My education level influences my online shopping preferences.	4%	17%	27%	26%	26%
My income level influences my spending on Online Fashion Retail platform.	6%	12%	26%	35%	21%
My demographic characteristics influence my purchasing decisions on Online Fashion Retail platform.	8%	11%	21%	31%	29%
My location influences the products I purchase on Online Fashion Retail platform.	6%	14%	21%	29%	30%
The availability of delivery services in my area influences my purchases on Online Fashion Retail platform.	6%	18%	23%	36%	17%
Regional fashion preferences influence my product choices on Online Fashion Retail platform.	7%	11%	23%	31%	28%
Local climate influences the clothing products I purchase on Online Fashion Retail platform.	6%	13%	23%	31%	27%

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Statement	SD	D	N	A	SA
Geographical factors influence my purchasing pattern on Online Fashion Retail platform.	6%	13%	23%	34%	24%

Distribution of Independent Variable Responses (Demographic & Geographical Segmentation)



IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Analysis

Across the ten independent-variable statements, combined Agree and Strongly Agree responses average around 57%, with about 20% disagreeing overall. My location and my demographic characteristics show the strongest combined agreement, both around 59-60%. The availability of delivery services shows the weakest response, with only 17% Strongly Agree and 24% combined disagreement.

Interpretation

Respondents generally agree that personal and location-based factors shape their shopping choices on Online Fashion Retail platform. Location and demographic background stand out as particularly influential drivers. Delivery availability, however, appears to be a comparatively weaker influence.

Distribution of Dependent Variable Responses (%)

Customer Purchasing Pattern

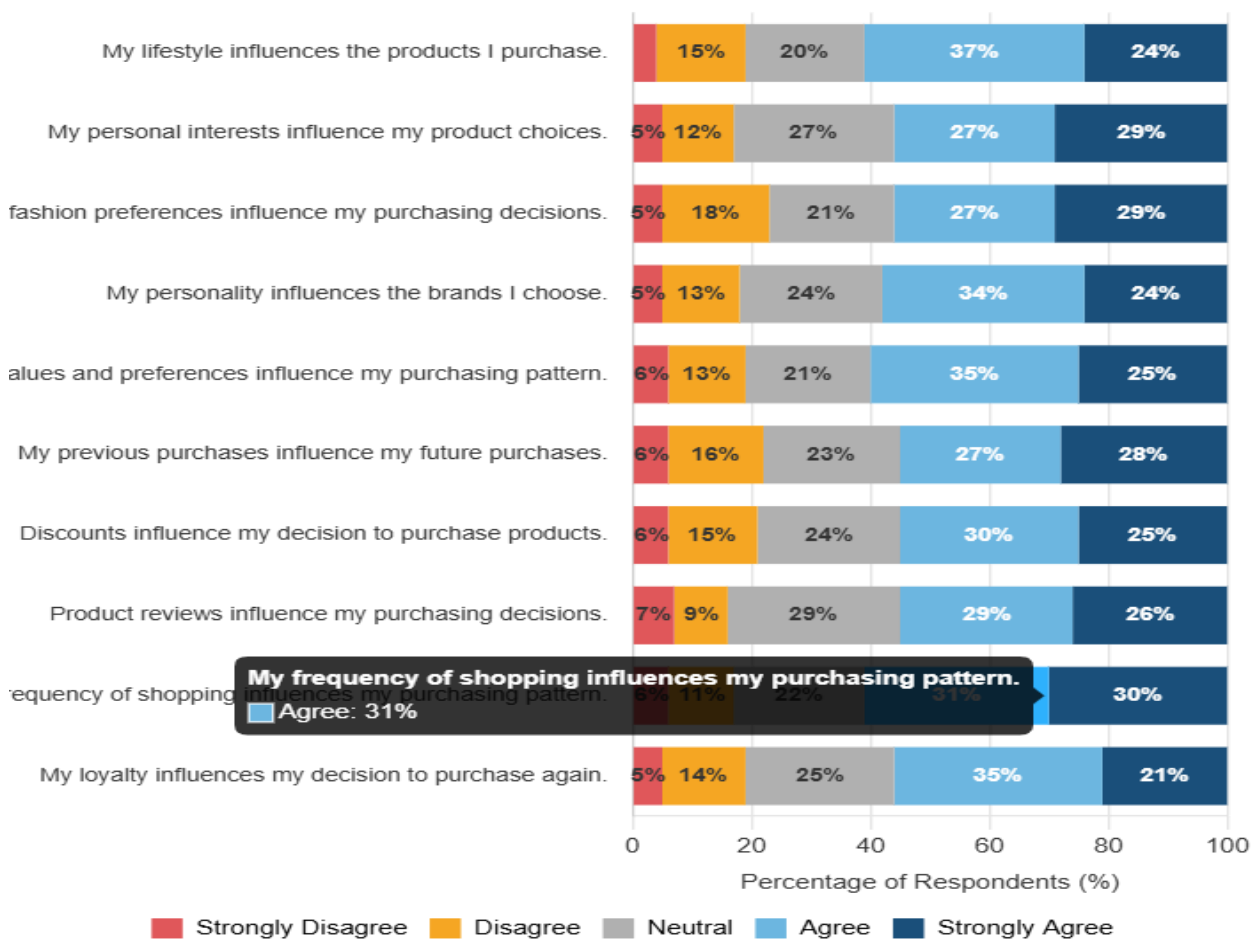
Statement	SD	D	N	A	SA
My lifestyle influences the products I purchase on Online Fashion Retail platform.	4%	15%	20%	37%	24%
My personal interests influence my product choices on Online Fashion Retail platform.	5%	12%	27%	27%	29%
My fashion preferences influence my purchasing decisions on Online Fashion Retail platform.	5%	18%	21%	27%	29%
My personality influences the brands I choose on Online Fashion Retail platform.	5%	13%	24%	34%	24%
My values and preferences influence my purchasing pattern on Online Fashion Retail platform.	6%	13%	21%	35%	25%
My previous purchases influence my future	6%	16%	23%	27%	28%

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**

Statement	SD	D	N	A	SA
purchases on Online Fashion Retail platform.					
Discounts influence my decision to purchase products on Online Fashion Retail platform.	6%	15%	24%	30%	25%
Product reviews influence my purchasing decisions on Online Fashion Retail platform.	7%	9%	29%	29%	26%
My frequency of shopping influences my purchasing pattern on Online Fashion Retail platform.	6%	11%	22%	31%	30%
My loyalty to Online Fashion Retail platform influences my decision to purchase again.	5%	14%	25%	35%	21%

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Distribution of Dependent Variable Responses (Psychographic & Behavioral Segmentation)



Analysis

Across the ten dependent-variable statements, combined Agree and Strongly Agree responses average around 57%, closely matching the independent variable block. My lifestyle and my frequency of shopping show the highest combined agreement, both around 61%. The two composite scores correlate very strongly, at approximately 0.98, showing they move together closely.

Interpretation

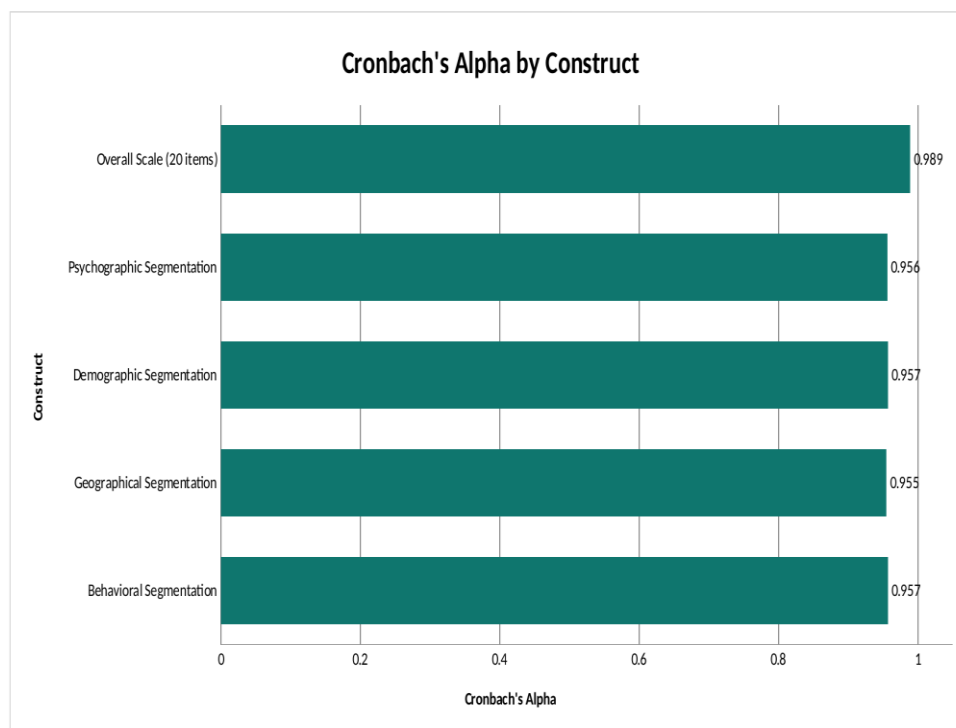
Respondents strongly link their purchasing pattern on Online Fashion Retail platform to lifestyle and shopping frequency. Product reviews are rarely dismissed outright, though many respondents

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

remain undecided about their influence. This pattern closely tracks the demographic and geographical pattern seen earlier.

Reliability Analysis (Cronbach's Alpha)

Construct / Scale	No. of Items	Cronbach's Alpha
Behavioural Segmentation	5	0.957
Geographical Segmentation	5	0.955
Demographic Segmentation	5	0.957
Psychographic Segmentation	5	0.956
Overall Scale	20	0.989



IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Analysis

The overall 20-item scale recorded a Cronbach's alpha of 0.989, and each of the four five-item constructs recorded an alpha between 0.955 and 0.957. All five alpha values exceed the commonly accepted minimum threshold of 0.70, and each construct-level value also clears the 0.90 mark generally considered excellent. Behavioral Segmentation and Demographic Segmentation recorded the joint-highest construct alpha of 0.957, while Geographical Segmentation recorded the lowest at 0.955.

Interpretation

The reliability score for the survey is excellent. All four constructs show strong internal consistency. This means the survey questions are reliable and measure what they are meant to measure. As a result, the findings drawn from this questionnaire can be trusted for further analysis.

related aspects of customer behaviour.

Hypothesis Testing Result

- **H₀ (Null Hypothesis):** Demographic, Geographical, and Psychographic segmentation do not have a significant influence on the behavioral Segmentation (Customer Purchasing Pattern) of Online Fashion Retail platform customers.
- **H₁ (Alternative Hypothesis):** Demographic, Geographical, and Psychographic segmentation do have a significant influence on the behavioral Segmentation (Customer Purchasing Pattern) of Online Fashion Retail platform customers.

Dependent Variable	R ²	Significant Predictor (p<0.05)	Hypothesis Result
Behavioral Segmentation (Customer Purchasing Pattern)	0.944	Demographic Segmentation (p < 0.001)	Accept H ₁
Behavioral Segmentation (Customer Purchasing Pattern)	0.944	Geographical Segmentation (p = 0.007)	Accept H ₁
Behavioral Segmentation	0.944	Psychographic Segmentation	Accept H ₁

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

(Customer Purchasing Pattern)		(p = 0.013)	
-------------------------------	--	-------------	--

Regression Analysis

DV IV	CP11		CP12		CP13		CP14		CP15		CP16		CP17		CP18		CP19		CP20	
R ²	0.803		0.848		0.831		0.819		0.865		0.837		0.816		0.782		0.864		0.773	
Predictor	E	P	E	P	E	P	E	P	E	P	E	P	E	P	E	P	E	P	E	P
CS 1	0.0238	.843	0.15297	.162	0.26039	.033	0.0761	.511	0.1157	.263	-	.413	0.00728	.953	0.2209	.094	0.2729	.010	0.0804	.530
											0.0969									
CS 2	0.0527	.652	0.07170	.501	0.33396	.005	0.2187	.056	0.1321	.192	0.0126	.913	0.12182	.309	0.1050	.413	-	.341	0.0322	.796
																	0.0977			
CS 3	0.2544	.018	0.07619	.429	0.29402	.007	0.0916	.372	-	.512	-	.527	0.02636	.807	-	.509	0.0467	.614	0.0748	.508
									0.0598		0.0661				0.0766					
CS 4	0.0298	.765	0.00851	.925	-	.942	-	.198	0.1543	.075	0.0436	.659	0.06900	.499	0.2235	.043	0.3075	<.001	0.1216	.256
					0.00737		0.1250													
CS 5	0.2362	.022	0.23321	.013	-	.186	0.0219	.824	-	.326	0.3474	<.001	-	.284	0.1008	.366	0.0270	.761	0.1227	.260
					0.13573				0.0861				0.11144							
CS 6	0.1328	.210	0.14811	.125	-	.518	0.1031	.314	0.0431	.636	0.1337	.201	0.10444	.334	0.0539	.641	-	.677	0.0294	.794
					0.06866												0.0384			
CS 7	0.0511	.626	0.09386	.327	-	.907	0.1303	.202	0.3399	<.001	0.1726	.099	0.18989	.079	0.1259	.275	0.0832	.366	0.1187	.292
					0.01232															
CS 8	0.0376	.714	0.14847	.115	-	.813	0.3054	.003	0.0922	.299	0.1991	.052	0.03709	.724	0.0430	.702	0.2028	.026	0.0125	.909
					0.02443															
CS 9	0.0743	.484	0.09778	.312	0.27595	.011	0.0374	.716	0.2205	.017	0.0629	.549	0.25680	.020	-	.667	0.1802	.055	0.1585	.164
															0.0499					
CS 10	0.0241	.805	-	.478	0.11231	.257	0.0713	.453	0.0609	.472	0.2061	.036	0.30506	.003	0.2147	.048	0.0316	.713	0.1670	.114
			0.06345																	

Analysis

The model explains 94.4% of the variance in behavioral Segmentation scores ($R^2 = 0.944$) and is statistically significant overall ($F(3, 96) = 534.737, p < .001$). Of the three predictors, Geographical Segmentation, Demographic Segmentation and Psychographic Segmentation all emerged as

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

statistically significant predictors at the 5% level. Demographic Segmentation had the strongest effect, with an unstandardized coefficient of $B = 0.446$ ($p < .001$), followed by Geographical Segmentation at $B = 0.289$ ($p = 0.007$) and Psychographic Segmentation at $B = 0.251$ ($p = 0.013$). The intercept of 0.038 was not statistically significant ($p = 0.684$), indicating no meaningful baseline effect when all predictors are zero. The adjusted R^2 of 0.942 is nearly identical to the R^2 value, showing that the model's explanatory power is not inflated by the number of predictors used. Together, these results indicate that behavioral Segmentation is very strongly predicted by the other three constructs.

Interpretation

The model explains a large part of the change in behavioural Segmentation scores. Geographical, Demographic, and Psychographic Segmentation all have a positive effect on it. This shows a strong link between these constructs and Behavioural Segmentation. Demographic factors appear to have the greatest influence, followed by geographical and psychographic factors.

Table: Correlation Matrix — Demographic and Geographical Items

Item	CPP1	CPP2	CPP3	CPP4	CPP5	CPP6	CPP7	CPP8	CPP9	CPP10
CS1	1.00	0.86	0.80	0.80	0.87	0.87	0.83	0.80	0.84	0.83
CS2	0.86	1.00	0.78	0.83	0.81	0.85	0.85	0.84	0.80	0.77
CS3	0.80	0.78	1.00	0.80	0.83	0.78	0.81	0.83	0.83	0.84
CS4	0.80	0.83	0.80	1.00	0.80	0.80	0.78	0.82	0.81	0.74
CS5	0.87	0.81	0.83	0.80	1.00	0.82	0.81	0.80	0.84	0.81
CS6	0.87	0.85	0.78	0.80	0.82	1.00	0.82	0.81	0.85	0.79
CS7	0.83	0.85	0.81	0.78	0.81	0.82	1.00	0.84	0.81	0.79
CS8	0.80	0.84	0.83	0.82	0.80	0.81	0.84	1.00	0.82	0.80
CS9	0.84	0.80	0.83	0.81	0.84	0.85	0.81	0.82	1.00	0.80
CS10	0.83	0.77	0.84	0.74	0.81	0.79	0.79	0.80	0.80	1.00

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Correlation Heatmap

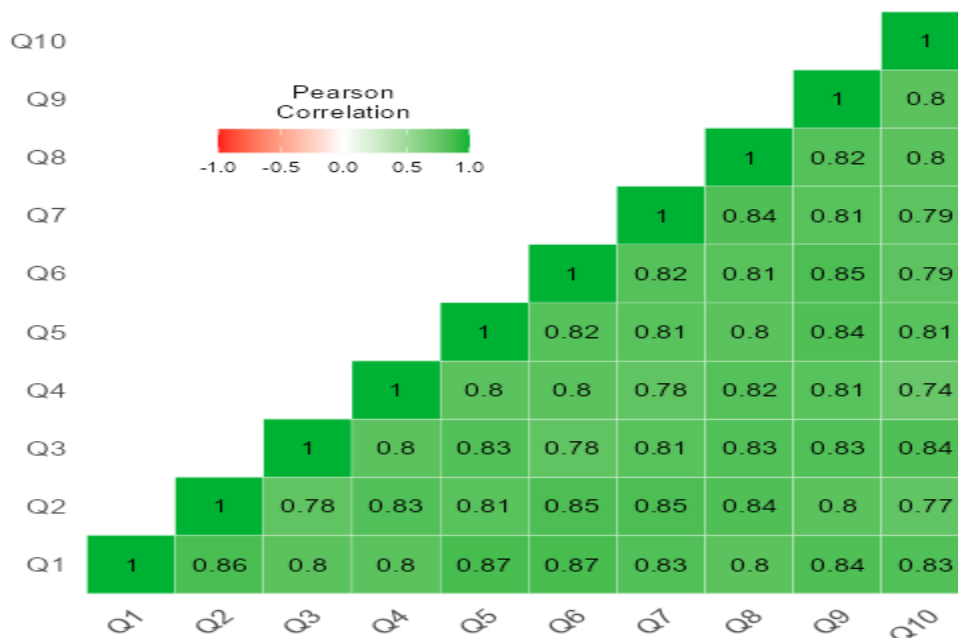
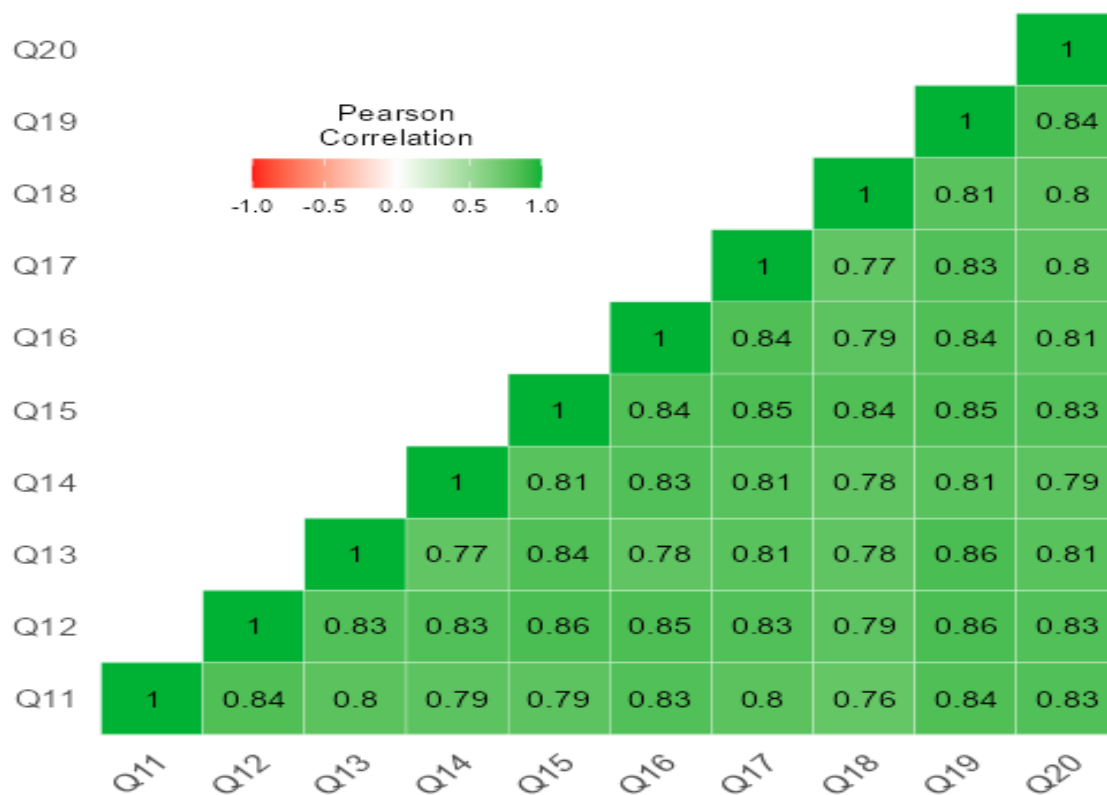


Table: Correlation Matrix — Psychographic and behavioral Items (Q11–Q20)

Item	CP11	CP12	CP13	CP14	CP15	CP16	CP17	CP18	CP19	CP20
CS11	1.00	0.84	0.80	0.79	0.79	0.83	0.80	0.76	0.84	0.83
CS12	0.84	1.00	0.83	0.83	0.86	0.85	0.83	0.79	0.86	0.83
Cs13	0.80	0.83	1.00	0.77	0.84	0.78	0.81	0.78	0.86	0.81
CS14	0.79	0.83	0.77	1.00	0.81	0.83	0.81	0.78	0.81	0.79
CS15	0.79	0.86	0.84	0.81	1.00	0.84	0.85	0.84	0.85	0.83
CS16	0.83	0.85	0.78	0.83	0.84	1.00	0.84	0.79	0.84	0.81
CS17	0.80	0.83	0.81	0.81	0.85	0.84	1.00	0.77	0.83	0.80
CS18	0.76	0.79	0.78	0.78	0.84	0.79	0.77	1.00	0.81	0.80

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Item	CP11	CP12	CP13	CP14	CP15	CP16	CP17	CP18	CP19	CP20
CS19	0.84	0.86	0.86	0.81	0.85	0.84	0.83	0.81	1.00	0.84
Cs20	0.83	0.83	0.81	0.79	0.83	0.81	0.80	0.80	0.84	1.00

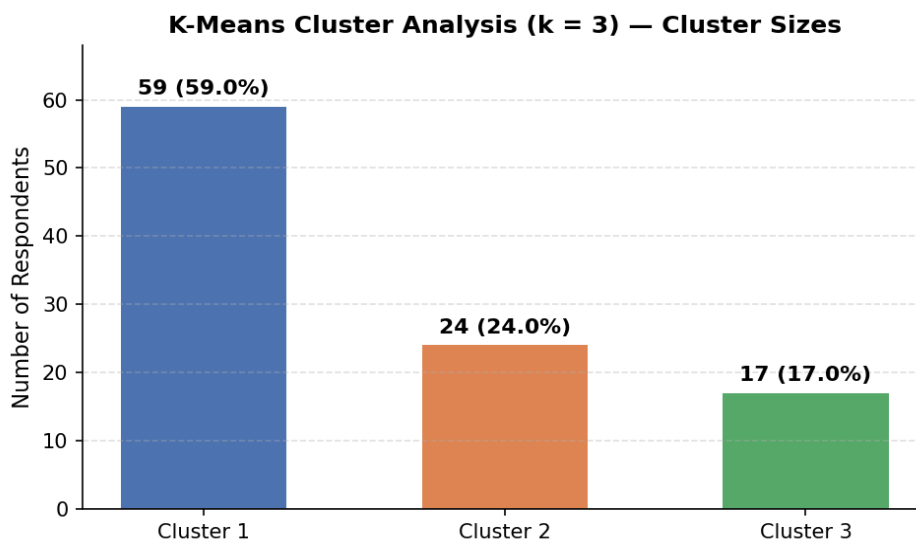


Correlation Heatmap

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Cluster Analysis (K-Means, k = 3)

Cluster	No. of Respondents	Percentage of Sample	Cumulative Frequency	Cumulative Percentage
Cluster 1	59	59.0%	59	59.0%
Cluster 2	24	24.0%	83	83.0%
Cluster 3	17	17.0%	100	100.0%



Analysis

A K-means procedure (k = 3) was run on the standardised responses to all 20 questionnaire items and assigned 59 respondents (59.0%) to Cluster 1, making it the largest segment. Cluster 2 contained 24 respondents (24.0%), the second-largest group, and Cluster 3 was the smallest, containing 17 respondents (17.0%). The gap between the largest and smallest clusters is 42 percentage points, showing a fairly uneven split in cluster sizes. Cluster 1 alone accounts for more than half of the

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

total sample, while Clusters 2 and 3 together make up the remaining 41 respondents (41.0%). Examining the mean item scores by cluster shows a clear separation in overall sentiment: Cluster 1 averaged 4.33–4.38 across the Demographic, Geographical, Psychographic and Behavioural constructs, indicating consistently strong agreement; Cluster 2 averaged 2.94–3.05, reflecting a moderate, near-neutral stance; and Cluster 3 averaged 1.71–1.85, reflecting consistent disagreement. A principal component projection of the 20 items shows the three clusters occupying clearly separated regions along the first component, which alone explains 82.7% of the variance in responses, confirming that the three-cluster solution captures a genuine and well-separated segmentation rather than an arbitrary split.

Interpretation

Cluster 1, the largest group, represents highly satisfied and engaged customers who responded positively across all four segmentation dimensions. Cluster 2 represents a moderate, fence-sitting group whose views sit close to neutral. Cluster 3, the smallest group, represents a critical or disengaged segment whose responses were consistently negative. This shows that respondents fall into three meaningfully different customer types rather than a single uniform group. This grouping can help Online Fashion Retail platform design different strategies for each type of customer, such as retention-focused engagement for Cluster 1, targeted persuasion for Cluster 2, and root-cause investigation or win-back efforts for Cluster 3.

Customer Segmentation Analysis – Online Fashion Retail Platform, Tumkur

Overall

This dashboard presents a consolidated view of the demographic segmentation survey conducted among 33 customers of an online fashion retail platform in Tumkur. It brings together the average scores recorded across the four segmentation dimensions — demographic, geographical, behavioural, and psychographic — alongside a breakdown of respondents by gender, age group, income, education, and shopping frequency. Interactive filters for gender, education, and age are provided on the right, allowing the data to be sliced by any of these categories. The visual layout combines summary cards at the top with bar charts and donut charts below, giving both a quick snapshot and a detailed distribution

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

view. Together, these visuals offer a single-page profile of who the surveyed customers are and how their scores compare across segments.



Analysis

The dashboard is based on responses from 100 customers, with average scores of 3.56 for demographic segmentation, 3.57 for geographical segmentation, and 3.57 for psychographic segmentation. Gender distribution shows Male respondents leading at 36%, followed by Other at 35% and Female at 29%. The 35–44 age group is the largest at 33%, while the ₹40,001–₹60,000 and Above ₹60,000 income brackets are tied at 28% each. Postgraduates form the largest education group at 28%, and shopping frequency peaks at 26% for customers who shop once a month.

Interpretation

The dashboard shows a customer base that is fairly evenly spread across genders, with no single category dominating overwhelmingly. Most respondents fall into the middle-adult age range, suggesting the core audience for online fashion e-commerce in this region skews toward established, working-age shoppers rather than the very young or older segments. Income and education levels lean toward the

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

higher end, pointing to a financially comfortable and well-educated customer base. Shopping habits reflect a mix of occasional and regular buyers, though monthly shoppers form the largest share, indicating steady rather than impulsive engagement with these platforms.

Dashboard 2: Customer Purchasing Behaviour Dashboard – Online Fashion Retail Platform, Tumkur

Overall

This dashboard focuses on how customer purchasing behaviour relates to shopping frequency for the online fashion retail platform audience in Tumkur. It cross-tabulates demographic and psychographic segmentation scores against shopping frequency to reveal whether stronger segmentation scores align with more frequent shopping. The purchasing frequency distribution chart summarises how often customers shop, while the two segmentation-by-frequency charts break this down further by score level. A horizontal bar chart on the right restates the shopping frequency categories as proportions, making the overall pattern easy to compare at a glance. This view draws on the full 100-respondent base rather than the 33-respondent filtered segment shown in the first dashboard.



IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Analysis

Among the 100 respondents in this view, 26% shop once a month, 21% shop more than five times a month, 21% shop rarely, 19% shop four to five times a month, and 13% shop two to three times a month. Combining all categories except "Rarely" shows that 79% of customers shop at least once a month or more. The psychographic segmentation chart shows the highest counts concentrated at score levels 4 and 5, peaking at 14 and 13 respondents respectively. The demographic segmentation chart follows a similar pattern, with its highest bar reaching 12 respondents at the top score band.

Interpretation

The results indicate that most customers shop on a fairly regular basis rather than only occasionally, with truly infrequent shoppers forming a smaller share of the base. Customers with stronger psychographic and demographic segmentation scores tend to cluster among more frequent shoppers, suggesting that lifestyle and personal characteristics play a meaningful role in shaping how often someone buys from this platform.

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**

CHAPTER 5

FINDINGS, SUGGESTIONS AND CONCLUSION

Findings

- The 35–44 age group is the largest segment (33%), with those 25+ making up 80% of respondents.
- Male respondents lead at 36%, followed by Other (35%) and Female (29%).
- Postgraduates form the largest education group (28%); Diploma holders the smallest (22%).
- 56% of respondents earn above ₹40,000 a month, with two income brackets tied at 28% each.
- 79% of respondents shop at least once a month, with monthly shoppers the largest single group (26%).
- Demographic and geographic segmentation statements show 57% average agreement, led by location and demographics (59–60%).
- Psychographic and behavioural statements also average 57% agreement, led by lifestyle and shopping frequency (~61%).
- The 20-item scale recorded an excellent Cronbach's alpha of 0.989, with all constructs above 0.95.
- Regression shows segmentation explains 94.4% of the variance in purchasing behaviour ($R^2 = 0.944$), with demographic segmentation the strongest predictor ($B = 0.446$).
- Correlation analysis shows a very strong relationship (~0.98) between segmentation and purchasing pattern scores.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Suggestions

- Weighting marketing campaigns toward the established, working-age customer base while still nurturing engagement among younger shoppers could help sustain broad appeal across age groups.
- Keeping product ranges and promotional messaging inclusive across all gender categories, rather than favoring any single group, would likely support a sense of fairness among a diverse customer base.
- Offering more detailed, informed product descriptions for highly educated customers, alongside simpler navigation for those with foundational qualifications, could make the platform more accessible to everyone.
- Expanding premium and mid-range product lines while continuing to maintain value-for-money options could serve the full spread of income groups represented among customers.
- Timing promotional pushes and flash sales around typical monthly spending cycles could resonate with the platform's most common shopping rhythm, while gentle reminders might better serve occasional shoppers.
- Since location and demographic traits stand out as strong influences on purchasing choices, tailoring recommendations around these characteristics could deepen the connection customers feel with the platform.
- Strengthening delivery speed, transparency, and communication could turn a currently weaker influence into a more positive driver of purchase decisions.
- Treating demographic, geographic, psychographic, and behavioral insights as parts of one connected system, rather than separate silos, could make marketing efforts feel more cohesive and consistent.
- Designing distinct engagement approaches for highly satisfied customers, moderately engaged customers, and more critical or disengaged customers could help address the differing needs within each group.
- Continuing to use the same well-performing questionnaire for future tracking studies could offer a reliable way of monitoring how customer segments and preferences evolve over time.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Conclusion

Study set out to examine the impact of customer segmentation on customer purchasing patterns using data modelling, with specific reference to customers of an online fashion-based e-commerce platform in Tumkur. The research was undertaken against the backdrop of a rapidly expanding digital retail environment in India, where increasing internet penetration, smartphone adoption, and improved logistics have transformed how consumers discover, evaluate, and purchase fashion products. Within this context, the study sought to move beyond a generalized view of the customer base and instead examine how demographic, geographic, psychographic, and behavioural characteristics shape the way customers engage with an online fashion platform. The demographic profile of respondents revealed a customer base concentrated in the 25–44 age range, indicating that established, working-age individuals form the core of the platform's audience rather than very young or older shoppers. Gender representation was found to be fairly balanced, with no single category overwhelmingly dominant, suggesting that the platform's appeal cuts across gender lines. Educational qualifications skewed toward postgraduate level, and a majority of respondents reported monthly incomes above ₹40,000, together painting a picture of a financially comfortable and well-educated customer segment. Shopping frequency data further showed that most customers engage with the platform at least once a month, reflecting a base that is actively and regularly involved in online fashion shopping rather than one that shops only occasionally or impulsively.

Core analytical contribution of the study lies in its examination of the relationship between segmentation and purchasing behaviour. All four segmentation dimensions — demographic, geographic, psychographic, and behavioural — recorded strong levels of agreement among respondents, generally falling between 57% and 61% combined agreement across the statements measured. This consistency across dimensions suggests that customers do not evaluate their purchasing decisions through a single lens but instead weigh a combination of personal, locational, lifestyle-based, and habitual factors simultaneously. The reliability analysis lent strong statistical credibility to these findings, with the overall scale achieving a Cronbach's alpha of 0.989 and each individual construct exceeding 0.95 — well above the threshold typically considered acceptable in social science research.

IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN TUMKUR

Reference

[Google Scholar](#) — Useful for finding research papers

. [ScienceDirect](#) — Strong source for e-commerce, consumer behaviour, marketing analytics, and data mining research.

[Springer Nature](#) — Useful for customer segmentation, machine learning, data analytics, and e-commerce studies.

[Wiley Online Library](#)

[Emerald Insight](#)

Books

1. Kotler, P., Keller, K. L., & Chernev, A. — Marketing Management
2. Laudon, K. C., & Traver, C. G. — E-commerce: Business, Technology, Society
3. Han, J., Kamber, M., & Pei, J. — Data Mining: Concepts and Techniques
4. Provost, F., & Fawcett, T. — Data Science for Business
5. Schiffman, L. G., & Wisenblit, J. — Consumer Behavior

APPENDIX

Appendix I: Structured Questionnaire

Demographic Profile

1. What is your age group?

☐ Below 18 years

☐ 18–25 years

☐ 26–35 years

☐ 36–45 years

☐ Above 45 years

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**

2. What is your gender?

☐ Male

☐ Female

☐ Other

☐ Prefer not to say

3. What is your highest educational qualification?

☐ PUC / Higher Secondary

☐ Diploma

☐ Undergraduate

☐ Postgraduate

☐ Other

4. What is your monthly income?

☐ Below ₹20,000

☐ ₹20,001–₹40,000

☐ ₹40,001–₹60,000

☐ ₹60,001–₹80,000

☐ Above ₹80,000

5. How often do you purchase products from Online fashion-based e-commerce platform?

☐ Very frequently

☐ Frequently

☐ Occasionally

☐ Rarely

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**

☐ Never

Demographic Segmentation

6. My age influences the types of products I purchase on Online fashion-based e-commerce platform. ☐

1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

My gender influences my product preferences on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

7. My education level influences my online shopping preferences.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

8. My income level influences my spending on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

9. My demographic characteristics influence my purchasing decisions on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

B. Geographical Segmentation

11. My location influences the products I purchase on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

12. The availability of delivery services in my area influences my purchases on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

13. Regional fashion preferences influence my product choices on Online fashion-based e-commerce platform

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

14. Local climate influences the clothing products I purchase on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

15. Geographical factors influence my purchasing pattern on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**

☐ 5 – Strongly Agree

C. Psychographic Segmentation

16. My lifestyle influences the products I purchase on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

17. My personal interests influence my product choices on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

18. My fashion preferences influence my purchasing decisions on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

19. My personality influences the brands I choose on Online fashion-based e-commerce platform.

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

20. My values and preferences influence my purchasing pattern on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

D. Behavioural Segmentation

21. My previous purchases influence my future purchases on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

22. Discounts influence my decision to purchase products on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

23. Product reviews influence my purchasing decisions on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

24. My frequency of shopping influences my purchasing pattern on Online fashion-based e-commerce platform.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

25. My loyalty to Online fashion-based e-commerce platform influences my decision to purchase again.

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**IMPACT OF CUSTOMER SEGMENTATION ON CUSTOMER PURCHASING PATTERN USING
DATA MODELLING IN ONLINE FASHION BASED E COMMERCE PLATFORM USERS IN
TUMKUR**